The Market for Part-Time Work and Maternal Participation in the Workforce

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Abstract

Women are over-represented in industries that feature shorter hours. This paper sheds light on the firm-side forces that give rise to this sorting. We study an immigration reform that markedly expanded the pool of commuting full-time workers in a pre-defined set of Swiss border localities using social security registers and business census data in a difference-in-differences design. We find that mothers — a group that predominantly works part-time — exit the workforce following the reform. Fathers, prime-age men, and women without children remain unaffected. We provide evidence that maternal drop-out is driven by a demand-side restructuring towards positions with longer hours: local employment shifts toward full-time positions, and job postings exhibit reduced demand for part-time work. Taken together, these findings suggest that firms dynamically adjust non-monetary aspects of jobs in response to supply, with reduced hours constituting an amenity that on average is costly to provide.

Keywords: Maternal Labor Supply, Non-wage amenities, Labor Demand, Part-time jobs
JEL: J16, J22, J23, J32

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1 Introduction

Women often work part-time. Across OECD countries, 23% of employed women work less than 30 hours per week compared to 8% of men. Jobs with lower average working hours also tend to cluster in industries that are generally female-dominated. Recent empirical work has documented that women's time constraints as mothers give rise to this pattern on the labor supply side, thus underlining the importance of lower-hours jobs in integrating maternal talent in the labor market (Wasserman, 2023; Ciasullo and Uccioli, 2023; Carry, 2026). However, to date we lack an understanding of the forces on the labor demand side that may either reinforce or weaken such gender-based sorting across industries. In particular, we have little insight into whether assigning tasks performed in a full-time job across more than one worker is costly to a firm, such that employers may prefer to hire full-time talent rather than mothers who work fewer hours.

Understanding potential demand-side constraints is important: If firms face costs of coordinating tasks across workers — essentially making production technologies convex in hours worked in the spirit of Goldin (2014) — then the reduced hours that keep mothers in the workforce make maternal employment costly to accommodate in broad parts of the labor market. However, how firms structure jobs has largely been treated as a fixed feature of production rather than a choice (see, e.g., Lachowska et al., 2026), in part because real-world settings to study such decisions are rare: tracing out firms' response requires a labor supply shift that is both plausibly exogenous and large enough to trigger meaningful adjustment.

In this paper, we use a large-scale but highly localized immigration shock of full-time workers in Switzerland to trace out both the supply and demand response in local labor markets. Relaxed regulations for cross-border workers (CBW) drastically increased the supply of commuting full-time workers from neighboring countries in Swiss border municipalities in the early 2000s. Using social security registers and a difference-in-differences design, we document that this reform led to a stark drop in workforce participation among Swiss mothers, a group that primarily works part-time when returning to the labor market after childbirth. In contrast, as well as men and women without children remain largely unaffected. Adding data covering the universe of Swiss establishments and a representative sample of job postings, we present several pieces of evidence that the main mechanism behind maternal dropout is reduced demand for part-time workers from the firm side. The patterns further point to low hours, rather than motherhood per se, as the reason mothers are substituted out: mothers who return to work full-time are unaffected by the reform, while those who would have returned part-time exit the workforce.

We study these questions in Switzerland, a country with relatively conservative gender norms and one of the largest "child penalties" in earnings that arises mostly due to part-time work, defined as working less than 90% of a full-time equivalent (FTE) (BFS, 2016). Swiss mothers' participation in the labor force increases steadily to up to 80% of mothers working from when their youngest child is between 0 and 16 years old. The large majority of returning mothers (80%) enter the

workforce part-time over this time horizon. At the same time, around 70% of part-time workers in the economy are mothers. Part-time work thus constitutes an important facilitator that allows mothers to re-enter the workforce after having children, especially since childcare beyond limited public school hours was not widely available during our study period.

We start by documenting the influx of foreign workers in Swiss local labor markets. As the reform raises the supply of CBW in municipalities within 15 driving minutes of the border, we adopt a donut design that compares these places to control municipalities more than 30 minutes away. An event study around the timing of the reform shows an expansion in the local share non-native workers by 7.2 ppt in treated municipalities — a 21% increase over the mean pre-reform, which is driven by full-time workers. The reform is unlikely to have allowed firms to simply hire cheaper labor: Consistent with legal guardrails ensuring the same working conditions and remuneration for CBWs, we show that post-reform hourly wage gaps between natives and CBW are precisely estimated and minimal (1.4%).

How are mothers — the main group in the labor market that works part-time — impacted when firms are able to access a larger pool of full-time talent? Using rich administrative data linking census and social security records of natives from 1994 to 2010, we focus on prime age workers with children below the age of 16, the time horizon over which mothers re-enter the workforce. We document that mothers' workforce participation in treated municipalities starts to decline following the onset of the reform. By 2010, mothers reduce their participation by 5.2 ppt, or 9% over the pre-reform mean. In contrast, we find no participation effects on fathers, nor on prime-age Swiss men or women who do not have children at baseline. This differential effect of the reform on mothers but not fathers remains when we compare individuals working in the same industry or occupation, and can thus not be solely explained by differential exposure to immigration by gender. Consistent with this, we show with survey data collected from firms that female workers in treated regions work more hours and are more likely to work full-time following the reform.

Why does the expansion in the supply of full-time workers push part-time mothers out of the workforce? We combine four complementary pieces of evidence to show that maternal drop-out reflects firms restructuring their workforce towards full-time positions — indicating that shorter hours are an amenity that is costly to provide. We start by documenting changes in the structure of jobs in treated local labor markets. Using data from the universe of Swiss establishments, event studies at the municipal level show that the composition of the local workforce becomes more full-time dominated: While the labor market and full-time jobs in the treated region expand, the share of jobs that are part-time declines by 3.9% relative to the pre-reform baseline. Disentangling these effects by nationality and gender highlights that the increase in full-time jobs is driven exclusively by non-Swiss workers. By contrast, the reduction in part-time jobs is borne by Swiss workers in low hours part-time jobs, and among them predominantly by Swiss women. In percentage terms, the low-hours workforce contracts by a similar amount among Swiss men and women, suggesting that firms shed low-hours jobs rather than female workers. Because part-time work was far more

prevalent among Swiss women to begin with, they bear most of this adjustment in absolute terms.

Second, we examine heterogeneous treatment effects by pre-reform labor market tightness to assess whether firms are more likely to change their labor mix if previously constrained: if part-time employment is a response to scarce full-time labor rather than a preferred arrangement, firms should adjust their workforce post-reform where that scarcity was more binding initially. We measure baseline slack as the number of unemployed mobile workers over all employed workers in a commuting zone by three-digit industry cell using data from the national unemployment registers. We find large compositional shifts towards full-time workers for establishments that had operated in local labor markets with above median tightness pre-reform. These compositional shifts are driven by increases in non-Swiss full-time workers and a decrease in Swiss female part-time employees. In contrast, the worker mix in establishments operating in slack labor markets remains unchanged.

We then correlate these compositional shifts with occupational characteristics considered in Goldin (2014) that may make accommodating part-time work more costly for firms: We observe larger reductions in part-time hours in industries whose occupations are more "greedy" and feature a higher degree of interdependence between coworkers via team work. In contrast, the association with occupational features that merely capture proximity in working with others is muted. This pattern is consistent with the cost of part-time work arising from coordinating interdependent tasks across workers.

Third, we measure firms' labor demand directly, using a representative sample of job advertisements posted by Swiss firms (Buchmann et al., 2022*a*). After the reform, establishments in treated regions become 8.2 ppt (9.2%) more likely to advertise a position as full- rather than part-time. Advertised positions also carry higher occupational prestige, indicating that firms may redefine roles rather than look for workers who would do the same types of tasks for more hours.

Fourth, we document heterogeneity in the social security data that corroborates that low hours are the main reason why mothers are substituted out. We proxy for part- vs. full-time work by comparing mothers' earnings after the birth of their first child relative to their pre-birth income. We classify mothers as working full-time if they recoup a large share ($\geq$ 90%) of their pre-birth earnings, and we validate this measure against hours reported in the Census. An event study shows that the composition of mothers in the workforce changes post-reform: The share of mothers who return to the labor market in a full-time capacity is unaffected, while the share returning part-time falls. These patterns indicate that mothers willing to return in a full-time capacity are unaffected by the inflow of full-time CBW, while mothers who would have returned to the workforce in a part-time capacity are instead staying out of the workforce entirely. In line with this, we show that maternal drop-out is mostly operating along the hiring margin: The reduction in maternal participation is stronger for mothers who were out of the workforce at the onset of the reform.

We rule out several alternative mechanisms that could also rationalize maternal dropout. We start by examining the role of wages. CBW might be cheaper to hire for firms in general, thus replacing mothers as a group whose labor supply is particularly elastic. We find little evidence that

would support this hypothesis: Consistent with the regulatory guardrails of the reform, the wage gap between CBW and natives shrinks to less than 1.5 log points when controlling for differences in observable characteristics and workplace. As a substantial share of CBW is male, CBW workers on average earn 3.8 log points more relative to Swiss women. Estimating reform impacts on hourly wages among natives who remain in the workforce further suggests limited scope for wage effects being the main driver behind maternal drop out.

We then explore whether higher household income post-reform could explain our findings. While partners of treated mothers experience somewhat higher earnings following the reform (+1% by 2010), the magnitude of this income effect cannot plausibly account for the full extent of mother's participation decline. Indeed, semi-elasticities for mothers' labor supply with respect to partner's earnings are two orders of magnitudes smaller than what would be needed to fully explain our main effect. We further rule out that increased fertility post-reform can account for our main treatment effects: While mothers who are younger at baseline are indeed more likely to have additional children post-reform, both younger and older mothers reduce their participation in the workforce.

Finally, we connect our results to the literature on the "child penalty" (Kleven, Landais, and Leite-Mariante, 2023) and estimate changes in the participation penalty as a consequence of the reform. In a triple difference-in-differences design, we compare how treated mothers' labor supply changes post-reform for each of the ten years following the birth of their first child. Mothers in treated regions experience a relative increase in the penalty of around 2 ppt, which is precisely estimated in the first five years after the birth and similar in magnitude but more imprecise in the years thereafter. Across all post-birth years, this constitutes an increase in the child penalty of around 6% relative to the average penalty in the control.

Our findings contribute to three main strands of literature. First, our study relates to work that seeks to understand which factors contribute to or cushion the decline in maternal labor supply and earnings (see, e.g., Cortés and Pan, 2023; Albanesi, Olivetti, and Petrongolo, 2023, for a review). Within this literature, an emerging strand has focused on the 'structure of work' in shaping such gender differences (Goldin, 2014; Goldin and Katz, 2016). Recent empirical work has provided compelling evidence that demand side regulation, such as hours requirements (Wasserman, 2023; Carry, 2026) or parents' rights to request flexible work hours (Ciasullo and Uccioli, 2023) are policy levers that shape female labor supply. Our study provides novel evidence on the demand side by showing that mothers drop out of the labor market once firms are able to access a wider pool of full-time workers. This highlights an unfortunate, but fundamental tension with respect to more flexibility on work hours, and emphasizes a broader need to better understand the 'structure of work' from the firm side when assessing policy tools to increase labor supply among workers with time constraints.

Second, we speak to an active literature in labor economics on job amenities and its relation to gender gaps across firms. Recent studies either use revealed preference approaches by examining worker moves across firms (Sorkin, 2018; Morchio and Moser, 2026; Lachowska et al., 2026) or

directly elicit workers' willingness to pay for specific aspects of jobs, such as the option to work part-time (Mas and Pallais, 2017; Wiswall and Zafar, 2018; Maestas et al., 2023; Caldwell, Haegele, and Heining, 2025; Humlum, Rasmussen, and Rose, 2025; Datta Gupta et al., 2025; Adrjan et al., 2026). Both strands share a focus on the labor supply side that treats firms, and their respective amenities, largely as fixed. Using a large-scale labor supply shock, we are able to fill in important aspects on the provision of amenities from the demand side: We show that firms may dynamically adjust the type of jobs offered, and our results are in line with reduced hours being an amenity that on average is costly to provide.

Third, we contribute to work on differential adjustments to immigration by gender. This literature has mostly examined low-skill shocks with immigrants affecting female workers differently because of specialization across industries and occupations (Amuedo-Dorantes and De La Rica, 2011; Edo and Toubal, 2017; Borjas and Edo, 2026), or because they relax native mothers' time constraints by supplying domestic services such as childcare (Cortes and Tessada, 2011; Cortes and Pan, 2013; Cortés and Pan, 2019; Cortés, 2023; East and Velásquez, 2024). We highlight an additional channel: immigrants' effect on the job amenities firms offer, in particular the option to work part-time. Competition in our setting thus operates along the hours margin. We also complement prior work on the firm-level effects of this reform. Using surveys on firms and workers, Beerli et al. (2021) document hourly wage gains for highly educated natives as firms overcome labor shortages and expand. Administrative registers covering the universe of the Swiss population allow us to examine distributional impacts at the extensive margin as firms adjust their input-mix (Lewis, 2011; Dustmann and Glitz, 2015). Consistent with prior work on immigration, groups that are less attached to the workforce are most at risk of displacement (Dustmann, Schönberg, and Stuhler, 2017, 2016).

This paper is organized as follows. The next section describes the setting and reform. Section 3 explains the data, empirical strategy, and provides descriptive statistics. Section 4 documents our main result and robustness. Section 5 provides evidence on mechanisms. Section 6 documents the reform's impact on the child penalty. Section 7 concludes.

2 Setting and Reform

2.1 Reform

In June 1999, the Swiss government signed a package of bilateral agreements with the European Union that included the "Agreements on the Free Movement of Persons" (AFMP), introducing free worker mobility between Switzerland and European countries. Because ratification required approval by the EU member states, the European Parliament, and in particular the Swiss electorate, it remained uncertain whether and when the agreement would take effect. Approval was granted in 2000, and the AFMP was enacted in June 2002, 1.5 years later than initially announced. Since Beerli et al. (2021) note that anticipatory effects are possible from the 1999 announcement onward,

and in particular after the Swiss popular referendum of May 2000, we follow their approach and define 1999 as the last pre-reform year.

The AFMP liberalized access for two types of workers: permanent immigrants with residency anywhere in Switzerland, and cross-border workers (CBW) who hold residency in a neighboring country (Germany, Austria, France, Italy) and commute to work in the Swiss *border region*. The border region was defined in bilateral agreements concluded between 1928 and 1973 and was unchanged by the reform (see Appendix Figure F1). Access for CBW was liberalized in two phases:

Transition (1999–2003) As described by Beerli et al. (2021), cantonal migration offices gained additional discretion over CBW work permit applications following the announcement in 1999. In 2002, several formal restrictions were abolished: new permits became valid for five rather than one year and were no longer tied to a specific job, and CBW were allowed to commute weekly rather than daily.

Free (2004–) June 2004 marks the free movement phase, with firms gaining unrestricted access to CBW as Switzerland dropped the prioritization of residents which had required firms to go through a lengthy approval process when hiring a CBW.[1] In 2004, hiring a CBW became as easy as hiring a Swiss native. Empirically, we will distinguish between a *Transition* phase covering the first removal of restrictions from 2000-2003, and a *Free* phase starting in 2004 following Beerli et al. (2021).

Access for permanent immigrants was liberalized on a similar timeline, but, unlike the CBW reform, uniformly across Switzerland. Resident immigrants to Switzerland had been subject to yearly quotas and prioritization of natives. In 2002, the quotas for immigrants from European countries were increased and in 2004, the prioritization of natives was abolished. In 2007, all remaining restrictions for immigrants from European countries were abolished.[2]

2.2 The Swiss Labor Market and Maternal Workforce Participation

Switzerland has relatively conservative gender norms around maternal labor supply and features one of the largest child penalties in earnings among industrialized nations (Kleven et al., 2019; Krapf, Roth, and Slotwinski, 2020). Maternity leave is short by European standards, and formal childcare beyond limited public school hours was not widely available during our study period (Felfe, Lechner, and Thiemann, 2016; Ravazzini, 2018). As a result, the average mother remains detached from the workforce for multiple years after childbirth (BFS, 2016).

Appendix Figure B1 documents parents' workforce participation by the age of their youngest child, using the 2000 Census. Almost all fathers participate full-time, irrespective of the age of the youngest child. Mothers, in contrast, increase their participation rate gradually from about 50% to 80% as the youngest child approaches age 16, when mandatory schooling ends. This increase stems almost entirely from entry into part-time positions: the share of mothers working full-time remains

[1] In particular, firms had to provide proof that they could not find "within an appropriate period of time" an equally qualified resident worker to fill a vacancy.

[2] Beerli et al. (2021) document a small increase of resident immigrants in the border region following the expansion of CBW and interpret this as crowding-in of immigrants as a direct consequence of the CBW reform.

at roughly 10% until the youngest child reaches school age and rises only gradually thereafter. Once the youngest child turns 16, participation plateaus, with moderate movement from part- to full-time work.

Mothers also account for the bulk of part-time employment in the Swiss economy. Appendix Figure B2 shows the composition of prime-age full- and part-time workers: close to 70% of part-time workers are mothers, against 10% of full-time workers.

3 Data, descriptive statistics, and empirical strategy

3.1 Data

We combine several data sources spanning the relevant time period:

Linked Census - Social Security (OASI) We use the population census in 2000 to create our main sample that we then link to the Swiss social security register from the Old-Age and Survivors' Insurance (OASI). The census contains information on individuals' main demographics, education, place of living, and household composition.[3] We assign individuals to treatment based on the municipality in which they live in the year 2000 and restrict the sample to Swiss nationals with a unique personal identifier.[4] For our main results on mothers and fathers, we follow prime age workers aged between 25 and 54 whose youngest child is younger than 16 in the year 2000. To measure partners' labor force outcomes, we link couples living in the same household with household identifiers at baseline. To analyze the number of children in a household that are born during the reform period, we further link to the the Swiss population registry (STATPOP) for 2012.

The OASI register is spell-based data for the universe of all individuals who are employed, self-employed, or who receive benefits in Switzerland between 1994–2010. For each individual and job, the data records one entry with information on start and end month, and associated earnings. In order to capture a meaningful dimension of workforce attachment that abstracts from sporadic episodes of participation, we define an individual as being in the workforce if they have real yearly earnings from wage or self-employment above 3,900 CHF (indexed in 2000), which corresponds to working for more than one month at the 10th percentile of the wage distribution. We add all income from wage employment spells in a given year to measure wage earnings. We drop the year 1998 in all of our main analyses with social security data since about 5% of records for 1998 are missing non-randomly (see Martínez, Saez, and Siegenthaler (2021)).

Business Census To study the reform's effects on the composition of the workforce in local labor markets, we use data from the Business Census, which is available for the years 1991, 1995,

[3] As uncertainty about whether the reform would be implemented was only resolved in May 2000 after passage of the popular referendum, we think it is reasonable to use the Census 2000 to measure baseline controls. For any parent, the decision to have a child will have taken place at least nine months prior.

[4] This identifier is created by the Federal Statistical Office via a probabilistic matching procedure and available for 94% of prime-age parents in the Census.

1998, 2001, 2005, and 2008 (Swiss Federal Statistical Office, n.d.*a*). The Business Census covers the universe of private and public establishments and reports establishment-level counts of workers by gender, nationality (Swiss, Non-Swiss), and employment level measured as a share of a full-time equivalent hours in the following categories: full-time ($\geq$ 90%), high part-time (50-89%), and low part-time (<50%). We assign workers to municipal labor markets based on the municipality of the establishment and use 1998 as the pre-reform baseline year. We drop agricultural establishments for all years, as data for this sector is not consistently available. As variables for 1991 are more limited, we can only include this year when examining jobs by full-time equivalent hours and restrict to establishments for which we can observe detailed industry information in a future year.

Swiss Job Market Monitor (SJMM) To measure labor demand, we use data from the Swiss Job Market Monitor for the years 1994-2010 (Buchmann et al., 2022*a*). The SJMM consists of a nationally representative sample of job vacancies posted by firms in Switzerland in the month of March and has been validated against official statistics on labor demand (Buchmann et al., 2022*b*). Starting in the early 2000s, the main sample collected from print media is supplemented with data from online media, company websites, and online job boards. The data contains rich information on the establishment advertising a particular vacancy (such as location and industry) and the nature of the job (employment level, occupation, desired applicant characteristics etc.). We categorize advertisements into full-time, high part-time and low part-time based on the employment level indicated in the job ad, and assign vacancies to treatment and control based on the location of the posting establishment. In addition, we use information on whether the advertised job includes managerial responsibilities and its occupational prestige based on the Standard International Occupational Prestige Scale (SIOPS) developed by Montgomery (1988).

Swiss Earnings Structure Survey (SESS) To examine hourly wages and hours, we rely on biennially collected data from the Swiss Earnings Structure Survey (SESS) for 1994-2010 (Swiss Federal Statistical Office, n.d.*b*). The survey is mandatory and comprises a large sample of private and public establishments with at least 3 employees in the secondary and tertiary sector. It collects detailed information on workers by enterprise (wage, hours, occupation, type of contract, etc.), but parental status is unobserved. We restrict the SESS sample to workers at prime age in 2000 who are on a monthly contract. This comprises 90% of all prime age workers and excludes workers hired at an irregular, hourly basis.

Unemployment Register (AVAM) To construct a measure of labor market tightness pre-reform, we use anonymized data from the national unemployment register provided by the Swiss State Secretariat of Economic Affairs for the years 1997-1999 (Swiss State Secretariat for Economic Affairs, n.d.). The data record all registered job seekers and their characteristics, including place of living and prior industry.

3.2 Descriptive Statistics

Maternal Participation in the Workforce – Appendix Table A1 presents summary statistics for our population of interest: Columns 1 and 5 refer to men and women who do not have children at the onset of the reform, whereas Columns 2 and 6 focus on parents. Parents have slightly more than two children at baseline, with the youngest child close to seven years old on average. Mothers are somewhat more likely than fathers or women without children to hold a secondary rather than a tertiary degree. Participation in the labor market is similar for men and women without children but differs sharply for parents. Industries are gender-segregated: around a third of mothers and women without children work in public administration, education or health (vs. around 14% of men), while men are more likely to work in manufacturing (around 18% versus 8% for women). Parents in the border and central regions are largely similar on demographics and fertility. Mothers in the border region are about 6 ppt more likely to hold a tertiary degree, work about two more weekly hours, and have higher median earnings, though participation rates are comparable to mothers in the Central region. Both mothers and fathers in the border region are somewhat more likely to work in knowledge-intensive services.

Inflow of CBW– Figure 1 documents patterns in the inflow of CBW from the SESS data that motivate our empirical strategy. We measure the municipal share of CBW as the number of CBW in a given year over the total number of workers in that municipality in 1998 as the last pre-reform year, so that compositional changes in the workforce are not conflated with general expansions or contractions of the local labor market. Figure 1 plots this share in bins of driving time to the nearest border crossing, with Panel (a) showing levels and Panel (b) showing changes relative to 1998.

Pre-reform, the level of CBW is substantially higher in municipalities within 15 driving minutes of the nearest crossing. While we observe a moderate increase in CBW with the relaxation of permit-specific restrictions from 2002 onwards, with the free phase starting in 2004, the share of CBW increases markedly through 2008. The increase is most pronounced in municipalities within 15 driving minutes of the border (Treated), considerably smaller in municipalities just beyond this threshold (Donut), and negligible beyond 30 minutes (Control). We will return to these patterns more formally by comparing Treated to Control municipalities in the main analysis in Section 4.

3.3 Empirical strategy

Following the descriptive patterns in Figure 1, we define municipalities as treated if they lie within 15 driving minutes of the nearest border crossing. Our main specification is the following event study with a donut design:

$$y_{imt} = \sum_{\substack{t=1994 \\ t \neq 1999}}^{2010} \gamma_t \cdot I_t \cdot Treat_m + \sum_{\substack{t=1994 \\ t \neq 1999}}^{2010} \eta_t \cdot I_t \cdot Donut_m + \alpha_m + \alpha_t + X_i \, \delta + \epsilon_{imt} \tag{1}$$

where y_{imt} is the outcome of interest for individual i in year t who lives in municipality m in the year 2000. I_t are indicator variables for a given year, and $Treat_m$ and $Donut_m$ denote whether an individual lived in a municipality within 15 minutes (Treat) or between 15 - 30 minutes (Donut) driving distance from the closest border crossing point at baseline. Year (α_t) and municipality (α_m) fixed effects control for common shocks and time-invariant municipality characteristics, respectively. We use data from the Census in 2000 to control for individual characteristics X_i at baseline.[5]

The coefficients of interest, γ_t, measure the impact of the reform for individuals who live in a treated municipality at baseline, relative to those in control municipalities beyond 30 minutes. The donut terms η_t absorb reform effects in intermediate municipalities, which are likely subject to spillovers. The key identifying assumption is that outcomes for individuals in treated and control municipalities would have evolved in parallel absent the reform. While this assumption cannot be tested directly, examining parallel trends in the pre-reform period allows us to assess its plausibility. We cluster standard errors at the commuting zone (CZ) level and estimate reform effects relative to 1999 as the last pre-reform year following Beerli et al. (2021).

We further report difference-in-differences (DiD) estimates that bundle the treatment impact relative to the pre-reform period during two periods of post-reform years:

$$\begin{aligned} y_{imt} = {} & \beta_1 \, Transition_t \cdot Treat_m + \beta_2 \, Free_t \cdot Treat_m \\ & + \eta_1 \, Transition_t \cdot Donut_m + \eta_2 \, Free_t \cdot Donut_m + \alpha_m + \alpha_t + X_i \delta + \epsilon_{imt} \end{aligned} \tag{2}$$

with $Transition_t$ an indicator for the transition period (2000-2003), and $Free_t$ an indicator for the fully liberalized period from 2004 onwards.

4 Main results: Reform Impact on Participation in the Workforce

4.1 Reform Impact on Non-Swiss Workers

We start by documenting the reform impacts on the presence of non-native workers more formally in an event-study using the Business Census. For each worker category, we measure the number of workers in a given municipality and year as a share of that municipality's total employment in 1998, the last pre-reform year. Holding the denominator fixed allows us to trace how different

[5] We measure all control variable and assign treatment status based on variables from the Census 2000, see Section 3.1. Individual level controls we include are: Age, age squared, highest level of education (primary, secondary, tertiary), bins for the age of the youngest child (0-3, 4-6, 7-12, 13-16, 17+) and bins for the total number of children (1, 2, 3, 4+).

pools of workers expand or contract relative to the pre-reform baseline without conflating these movements with changes in the size of local labor markets. We weight each municipal observation by 1998 employment, so that estimates are representative of the average worker.

Figure 2 documents a sharp increase in non-Swiss workers as a share of pre-reform local employment that amounts to 7.2 ppt by 2008.[6] This corresponds to an increase in non-native workers by 21% over the pre-reform mean (.34). While we can separately identify CBW in the Business Census only in a limited set of years, Appendix Figure C1 shows that the growth in non-Swiss workers is entirely accounted for by CBW for the years in which this information is available. Figure 2 also documents that most of the increase in non-native workers is coming from non-native full-time workers: By 2008, this group's share of the local labor market expands by 5.6 ppt or 20% over the pre-reform mean (.28).

4.2 Decline in Maternal Workforce Participation

Workforce Participation– Next, we document the reform's impact on parents' workforce participation. Figure 3 shows that the participation of mothers in treated municipalities evolves in parallel compared to mothers who live further away from the border until the onset of the reform in the early 2000s, and subsequently starts to decline. By 2010, participation rates of treated mothers are 5.2 ppt lower. This represents a 9% drop over the pre-reform mean (58.3%). Conversely, the reform did not meaningfully affect the workforce participation of fathers with yearly point estimates for later reform years indicating only a slight, and imprecisely measured decline.

Table 1 pools yearly estimates across the *Transition* and *Free* period (see Equation 2). Column 1 reports estimates with year and municipality fixed effects only, while Column 2 presents our preferred specification with individual level controls. Averaged over all years in the *Free* period, mothers' participation falls by 3.8 ppt in the *Free* period, while fathers' participation dips by 1.1 ppt. While precisely estimated when bundled across years, the drop for fathers is relatively small and amounts to a decrease of about 1% in this group's participation rate over the pre-reform mean (96.5%).

Earnings– Figure C2 reports the reform's impacts on yearly wage earnings, unconditional on employment. Consistent with the results on labor supply, mothers' earnings fall by 0.46 units of the inverse hyperbolic sine by 2010, while fathers' earnings are largely unaffected.[7]

Unemployment– Since our main outcome measures participation in the workforce, i.e. captures men and women who receive wage- or self-employment income, we separately examine reform impacts on unemployment benefit receipt in Appendix Figure C5. The outcome is an indicator equal to one if an individual received any unemployment benefits in a given year. We observe no

[6] Beerli et al. (2021) use survey data at the firm level from the SESS to estimate reform impacts on non-Swiss workers. Our point estimates from the Business Census coincide almost exactly with their estimates.

[7] Note that because our outcome has a mass point at zero and is positive otherwise, the inverse hyperbolic sine transformation places substantial weight on the extensive margin, and the coefficient should not be read as a percentage effect (Chen and Roth, 2024).

reform impact on fathers, and negative but noisy point estimates for mothers, consistent with this group being less likely to have a job and therefore less likely to qualify for unemployment assistance. Mothers who leave employment therefore do not appear as unemployed, indicating that the participation decline reflects exit from the labor force rather than a shift into unemployment.

Hours– Our main outcome is binary and therefore doesn't speak to how hours changed in treated labor markets. Since hours are unobserved in the social security data, we use survey data from the SESS to examine the impacts of the reform on hours worked conditional on employment. We restrict the sample to native workers in the same age group as our main analysis sample (prime age in 2000) but cannot separately identify mothers in these data. Appendix Figure C3 shows an increase in hours worked among treated women who are in the workforce in Panel (a) and that any given female worker in the treated region is more likely to be working full-time post-reform in Panel (b). As the SESS is not a panel, we cannot distinguish whether this reflects the exit of part-time women or hours increases among women who remain employed. We revisit this question in Mechanism Section 5.3, with evidence pointing to the former rather than the latter channel.

4.3 Robustness of Main Result

In the following, we document that motherhood — and not just being female — is the defining characteristic behind the participation decline of mothers. First, Appendix Figure C4 shows reform impacts on prime age men and women who do not have children at the onset of the reform. Prime age women without children at baseline do not experience a reduction in workforce participation. If anything, point estimates for this group are positive.[8] Patterns for prime age men without kids mirror those for fathers.

Second, we show that the decline in maternal participation is not accounted for by differential sorting of men and women across industries and occupations, which could have been differentially affected by the inflow of CBW. To do so, we compare mothers and fathers who work in the same industries or occupations by estimating our main specification with joint industry or occupation fixed effects measured at baseline, but allowing for separate treatment effects and controls by gender. One challenge when adding industry and occupation fixed effects is that these variables are more likely to be missing in the Census when an individual is not in the workforce. We therefore report two sets of estimates: i) controlling for a missing industry or occupation category, and ii) dropping individuals with missing values from the estimation sample. Column 1 in Appendix Table C1 repeats our main treatment effect. Coefficients are identical when we include and separately control for missing industry information in column 2 and quite similar when dropping individuals with missing industry information in column 3. A similar picture arises when including occupation fixed effects in columns 4 and 5. Column 6 and 7 report estimates when we include learned occupation fixed effects, which are available for a much larger share of the sample.

[8] 27% of this group of women end up having at least one child by 2010. See also Mechanism Section 5.5.3 for fertility impacts of the reform.

Additional robustness of our main result for mothers and fathers is reported in Table 1. Results remain similar when including individual fixed effects in Columns 3 and 9, NUTS-II trends as in the main specification of Beerli et al. (2021) (Columns 4 and 10), when controlling flexibly for differential time trends across municipalities by baseline maternal workforce participation (Columns 5 and 11), and when dropping large cities (Columns 6 and 12).

5 Mechanisms

Why does the expansion in the supply of full-time workers push part-time mothers out of the workforce? We present several pieces of empirical evidence that are consistent with a demand-side restructuring towards fewer part-time jobs in local labor markets driving the observed decline in mothers' workforce participation.

5.1 Structure of Jobs in Local Labor Markets

5.1.1 Compositional Changes

We start by examining shifts in the composition of the workforce in local labor markets with respect to full-time or part-time status using data on the universe of workers in the Business Census. For each establishment in Switzerland, these data entail the number of workers employed in jobs of three different hours categories (full-time, high part-time (50-89% of an FTE), and low part-time ($<$ 50% of an FTE)), by nationality (Swiss, Non-Swiss), and by gender.

As outcomes, we examine the share in each worker category and year at the municipal level by normalizing with total municipal workers in the last pre-reform year in the Business Census (1998). We weigh each municipal observation by total workers at baseline. This allows us to abstract from general expansions or contractions of the labor market such that we can interpret coefficients as increases or decreases in the respective job category relative to its baseline size following the reform.

To get a general sense of the overall job composition in the economy, the first column of Figure 4 depicts reform impacts on jobs by hours. We observe a (noisy) expansion in full-time jobs amounting to 4 ppt by 2008 in Panel 1a that is accompanied by a contraction of .06 ppt in low-hours part-time jobs ($<$ 50% of an FTE) in Panel 1c, and by a decline to a lesser extent in the high part-time category in Panel 1b.[9] Column 1 in Appendix Table D1 estimates pooled coefficients for the *Free* period for jobs in Panel A and hours contributed by different job types in Panel B, showing a (noisy) increase of 2.6 ppt (3 ppt) for full-time jobs (hours) and a precisely estimated decrease of 1 ppt (.6 ppt) for part-time jobs (hours), which amounts to a decline of 3.9% (4.2%) in that job category over the pre-reform mean.

The second column in Figure 4 illustrates the differential incidence of these shifts in the overall job composition by further differentiating the workforce by nationality and gender, which we can

[9] As documented in Appendix Figure D1, municipal labor markets experience an imprecisely estimated expansion in FTE of 3.7 ppt by 2008.

observe in the Business Census data starting in 1995. Panel 1b highlights that the growth in full-time workers is entirely accounted for by expansions in Non-Swiss workers from both genders, while the effects on natives working full-time are not statistically distinguishable from zero — suggesting that the native full-time workforce was already fully exhausted pre-reform. This contrasts with patterns observed for part-time jobs depicted in Panels 2 and 3, respectively. While the share of Non-Swiss workers also increases in "high-hours" part-time jobs, it does so to a much smaller extent compared to full-time positions (Panel 2b). For "low-hours" part-time jobs depicted in Panel 3, the share of part-time jobs declines both for Swiss women and men, and these contractions are not offset by increases in non-Swiss workers. In percentage terms, low-hours part-time jobs decrease to a similar extent both among Swiss women (-10%) and Swiss men (-9%) by 2008, consistent with a workforce restructuring along the hours rather than gender margin. As Swiss women in these low-hours part-time jobs represent a higher share of the workforce relative to Swiss men (7.3% vs 2.3%), they are more heavily impacted by these shifts in absolute terms.

We summarize these patterns for our main demographic groups of interest in Appendix Table D1: Increases in the non-Swiss full-time workforce amount to 3.4 ppt — a 19% increase over the pre-reform mean (Row 3). The Swiss female part-time workforce declines by 1.8 ppt or 11.5% over the pre-reform mean (Row 4). In Appendix Figure D2, we document that increases in non-native full-time and decreases in Swiss female part-time workers permeate most sectors of the economy following the reform: With the exception of construction, the share of non-Swiss workers increases in Services and Manufacturing, and to a lesser extent in Public Administration, Education and Health. We see in column 3 that all of these sectors experience increases in the share of non-Swiss full-time workers and decreases in the share of Swiss female part-time workers, which result in somewhat more noisily estimated shifts in the share of jobs that are filled by full- vs. part-time workers.

The estimates presented in this section so far take into account the entire labor market. To gauge to which extent these results may be driven by differential firm entry or exit following the reform, in column 2 of Appendix Table D1 we present estimates at the establishment level restricting the sample to firms that exist at baseline: For each establishment, we calculate the share of workers in each category and year relative to that establishment's size in the pre-reform year and estimate reform impacts including establishment fixed effects, weighting by establishment size at baseline. Reform impacts are unchanged and somewhat more precisely estimated, suggesting that established firms are indeed restructuring their workforce following the reform. In columns 3 and 4, we assess robustness of these results: Column 3 probes for potential SUTVA violations by showing that coefficients are unchanged when we exclude multi-establishment firms that operate both in the border and central region and may thus easily shift staff across different locations.[10]

[10]Note that it is unlikely that part-time women merely switch work location from the border to the central region: We show in the main results, which are based on a panel of natives for whom we assign treatment status based on pre-reform location, that mothers drop out of the workforce. Section 5.3 directly documents a shift from part-time to non-employment for mothers.

Column 4 documents that reform impacts are similar when excluding small establishments with less than 4 workers from the sample.

5.1.2 Heterogeneity by Labor Market Tightness

To further the notion that maternal drop-out may be the result of firms shifting their workforce composition from part- to full-time, we next examine heterogeneous treatment effects by the degree to which establishments in local labor markets were constrained in their hiring pre-reform. Firms that were most constrained in recruiting talent pre-reform were likely more willing to accommodate female workers on part-time schedules. Such firms should thus be more likely to re-optimize after the reform by substituting out part-time workers with full-time talent if part-time workers are less desirable. In contrast, firms that had no recruiting difficulties pre-reform should be less likely to exhibit such shifts.

To construct a proxy for local labor market tightness, we use data from the national unemployment registers. For each three-digit industry and commuting zone cell, we calculate the total average number of unemployed workers that are mobile, i.e. willing to commute, between the years 1997-1999.[11] For our measure of labor market tightness, we divide by the total number of workers in each industry-by-commuting-zone cell in 1998 as measured in the Business Census to obtain the average pre-reform share of workers that is actively looking for a job within each cell (see Appendix Figure D3). We define an establishment as operating in a tight labor market if the share of unemployed workers in its industry-by-commuting-zone cell is below the overall nationwide median (worker-weighted) at baseline.

Appendix Figure D4 shows establishment-level event studies. In Panels a and b, we first examine overall shifts in the full- vs. part-time composition of firms. In Panel (a), we see that establishments that operate in tight labor markets pre-reform experience a substantial increase in the share of full-time workers by 5.8 ppt in 2008. This amounts to an increase of 8% over the pre-reform mean. These increases are offset by a simultaneous reduction in part-time workers whose share of the workforce decreases by 2.6 ppt (or 9% over the pre-reform mean). In contrast, we observe no meaningful compositional shifts in establishments that were operating in pre-reform labor markets with plenty of slack in Panel (b). In Panels (c) and (d), we examine these patterns for our main groups of interest, non-Swiss full-time workers and Swiss female part-time workers. Indeed, the shifts observed among establishments in tight labor markets are driven by shifts in these particular groups (Panel c). While establishments in labor markets with slack also experience a small increase in non-Swiss full-time workers (Panel d), this does not result in an overall shift towards more full-time workers as these industries could likely obtain their desired labor-mix already pre-reform.

[11] 91.5% of all unemployed workers indicate willingness to commute. We use a crosswalk from the Swiss Federal Statistical Offices to link 3-digit industries in the Business Census (classification 2002) to 3-digit industries in the unemployment data, based on an industry classification from 1985. This results in a match for 94.1% of all establishments present at baseline, which form the sample for this part of the analysis.

5.1.3 Part-time Reductions and Job Characteristics across Industries

Are reductions in part-time hours following the reform concentrated in industries that feature higher costs to coordinate between workers? As a last exercise with the Business Census data, we examine differences in how much different industries reduce their reliance on part-time hours. We then relate these differences to industry-level job characteristics related to "greediness" considered in Goldin (2014), i.e. features that may contribute to make accommodating part-time work more costly for firms.

In Appendix Table D2, we report industry-specific reductions in part-time FTE as a share of total FTE at baseline. We restrict the sample for this exercise to establishments in tight labor markets pre-reform. We observe the largest reductions of hours performed by part-time workers in Health and Social Services as well as "High Skill" Services that span from architecture, engineering firms and software engineering firms to NGOs. Hospitals and Consulting industries exhibit somewhat smaller but still sizable declines, while Hospitality and "Low Skill" Services, such as cleaning and sanitation, experience an expansion in the share of hours contributed by part-time workers.[12]

To characterize the occupational features of these industries, we use the Census in 2000 to link each worker aged between 20 and 55 in Switzerland to O*NET occupation characteristics via their occupation code (U.S. Department of Labor, Employment and Training Administration, 2025). We then calculate worker weighted occupational characteristics by industry that reflect the occupational composition of Swiss industries at baseline. We construct three measures of interest by averaging across standardized O*NET scores: (i) Greediness, i.e. job features that correlate with higher residual gender earnings gaps among top-earnings occupations as depicted in Goldin (2014), ii) Interdependence and Teamwork, to capture to what extent a workers' tasks may be interdependent with others and therefore require costly coordination, and iii) Contact with Others, to proxy for job features that require social interactions without implying coordination costs.[13]

Appendix Figure D5 shows the relationship between the reduction in part-time hours and the occupational characteristics of each industry. We observe a strong negative association for occupational characteristics that relate to greediness, with part-time declines larger in industries where workers face more time pressure, more interactions with others and a higher degree of unstructured tasks in Panel (a). In Panel (b), we observe similarly that industries that feature occupations with a higher degree of team work and interdependence of decision-making are among those that reduce their reliance on part-time hours to a larger extent. Panel (c) underlines that these patterns are not just an artifact of women working in industries which are characterized by a higher level of social

[12] Appendix Table D3 shows the three largest detailed industries for these larger industry groups.

[13] The three respective measures average across the following standardized O*NET characteristics: Greediness: Contact with Others, Determine Tasks, Priorities and Goals, Frequency of Decision Making, Time Pressure, Establishing and Maintaining Interpersonal Relationships. Interdependence and Teamwork: Impact of Decisions on Co-workers or Company Results, Work With or Contribute to a Work Group or Team, Developing and Building Teams, Coordinating the Work and Activities of Others, Communicating with Supervisors, Peers, or Subordinates. Contact with Others: Contact with Others, Coordinate or Lead Others in Accomplishing Work Activities (not as a supervisor or team leader), Physical Proximity.

interactions: There is no strong association between workers having general contact with others and the reduction in part-time hours at the industry level in Panel (c). While purely correlational, these associations are consistent with reform impacts being concentrated in industries in which accommodating part-time talent pre-reform was particularly costly for firms.

Taken together, these three sets of results based on the Business Census are suggestive of larger shifts in the composition of the labor market with readily available full-time workers replacing part-time jobs. These shifts are concentrated in tighter labor markets where the reform more strongly relaxed hiring constraints, and in industries that are more heavily dominated by occupations that exhibit higher coordination costs between workers through interdependencies and team work.

5.2 Decline in Firms' Demand for Part-time Workers

To better assess the extent to which the above documented changes in local labor markets are driven by demand-side factors as opposed to supply, we continue by examining the reform's effects on firms' job posting behavior as a direct measure of labor demand. We use data from the Swiss Job Market Monitor (SJMM), a yearly representative sample of jobs advertised by Swiss firms. We classify jobs as a full-time or part-time position based on the percentage of an FTE indicated in the posting recorded by SJMM.

Figure 5 shows event study estimates with the outcome an indicator equal to one if the advertised job is part-time (i.e., less than 90% of an FTE). Due to the relatively smaller sample size, we estimate Equation 1 with year and large region (instead of municipality) fixed effects and add industry fixed effects. While individual yearly coefficients are not always statistically significant, we see a clear pattern towards firms becoming less likely to post a position as part-time post-reform. Table 2 pools coefficients across the transition and free period post-reform. In the *Free* period, the probability that a position is posted as full-time increases by 8.2 ppt or 9.2% over the pre-reform mean (column 1) and is born by similar sized reductions in the likelihood of an ad being for a higher or lower hours part-time position (columns 2 and 3). On average, the employment level of posted positions increases by 2.9 ppt. Columns 5–7 further examine the characteristics of posted positions to gauge the extent to which firms may re-organize production processes by recruiting for redefined roles as opposed to full-time workers performing the same, but just more of, a part-time worker's tasks. Column 5 shows suggestive evidence that posted positions are more likely to include managerial responsibilities, but this is not precisely estimated. Columns 6 and 7 document that posted positions exhibit higher occupational prestige based on the SIOPS scale by Montgomery (1988), and are more likely to be of upper-mid to high prestige.

5.3 Maternal Drop-out by Hours Supplied

As a last step, we return to the social security data to examine whether the patterns on the supply side of the labor market support the hypothesis that firms may substitute part-time female Swiss workers with full-time CBW due to a preference for higher-hours workers. Importantly, if hours

supplied is the dimension that matters for firms' choices, we should not observe negative labor market consequences for mothers who are willing to work full-time. While we cannot directly observe hours in the social security data, we construct a proxy thereof by using information on the share of pre-motherhood earnings (when most women work full-time) that women recoup upon returning to the workforce. If mothers recover an amount that is close to their pre-birth earnings, they are plausibly more likely to return to the labor market in a "full-time" capacity.

For this exercise, we restrict our main analysis sample to the post-birth years for women who had their first child after 1982 as the earliest year in which we can observe pre-birth earnings in the social security data. We focus on women who did not have earnings from self employment and were wage employed for all 12 months in the year before their first child is born ("pre-birth"). We convert earnings to real terms and trim pre-birth earnings at the 10th and 99th percentile for each year in the data and construct a "post-birth earnings share" by dividing real yearly wage earnings by pre-birth earnings. We trim the bottom 20% of the post-birth earnings share to exclude mothers who have a more intermittent labor force attachment from our employment measure. For the event studies, we define mothers as "full-time" employed workers if their post-birth wage earnings amount to at least 90% of their pre-birth value, and as "part-time" employed otherwise.

For the year 2000, we can validate this full-time proxy with data from the Census where mothers report whether they work full or part-time at the time of enumeration. In the sample of mothers with births after 1982, 15.62 % state in the Census that they work full-time (conditional on working), while this share is 15.28% based on our post-birth earnings measure. For mothers who report working full-time in the Census, post-birth earnings amount to 102% of their pre-birth wage. This is in contrast to mothers who report working part-time, who earn about 60% of their pre-birth earnings on average. In Appendix Figure D6, we plot the density of the share of pre-birth earnings by whether mothers report working full- vs. part-time. While the mapping is not perfect, the mass of each density is consistent with the reported hours in the Census.

Figure 6 documents our main event study on the sample of post-birth mothers. We observe no change in the likelihood of returning to the labor market in a full-time capacity, defined as recovering at least 90% of pre-birth earnings. The group of mothers who is willing to supply full-time hours thus remains unaffected by the CBW shock. In contrast, we observe a decrease in the likelihood that a treated mother is returning to the labor market in a part-time capacity. By 2010, the likelihood that a mother is working in part-time wage employment is about 5 ppt lower in the treated relative to the control areas. Defining an indicator equal to one if a mother is either full- or part-time employed based on these earnings measures, we observe a similar pattern and magnitude of drop-out in this post-birth sample relative to our main analysis sample. Appendix Table D4 reports the difference-in-differences coefficients and shows that these results are robust to different cut-offs to determine full-time capacity and to not trimming outliers. While our earnings-based measure of full- vs. part-time work is a proxy to categorize workers into categories of hours, these results nevertheless support the interpretation that firms substitute towards CBW primarily due

to (most) mothers' supply of lower hours.

5.4 Dropout at Hiring Margin

How does the reduction in part-time jobs take place? Two pieces of evidence from the social security data complement the job ads data and are suggestive of the impact operating along the hiring margin, i.e. particularly affecting mothers looking to return to the labor force after a break in participation. Appendix Table D5 splits our main sample of mothers by whether they were either in the workforce (columns 1–2) or in wage employment (columns 3–4) at baseline. The bulk of the decline in labor force participation is driven by mothers who do not work at the onset of the reform. Consistently, and as documented in Appendix Figure C4, women who do not (yet) have children at baseline and are attached to the labor force do not experience differential dropout, even though almost a third of this group (27%) will have given birth by 2010.

5.5 Alternative Mechanisms

5.5.1 Downward Pressure on Wages

While Switzerland put in place regulatory guardrails to ensure equal wages and working conditions for CBW, it is possible that firms may have tried to circumvent these restrictions. If the influx of CBW put downward pressure on native's wages, mothers may have found it less attractive to remain in the workforce. We present two pieces of evidence that suggest such an explanation is unlikely to account for the full impact on maternal drop-out.

First, we estimate gaps in log hourly wages between CBW and natives in the treated areas post-reform using the SESS data to understand whether CBW are a cheaper source of labor. Columns 1–4 in Appendix Table D6 document wage gaps gradually adding controls. We observe a noisily estimated raw gap in hourly wages between CBW and natives of -5.6 log points in column 1. Individual characteristics leave it largely unchanged but improve precision (column 2), while accounting for differences in industry and occupation narrows it to -2.7 log points (column 3). Once we compare CBW and natives who work in the same establishment, CBW earn only 1.4% less compared to natives (column 4). Overall, CBW thus do not constitute a substantially cheaper source of labor. This is even less so when directly compared to Swiss female workers in columns 5–8. Relative to female natives, CBW earn 3.8 log points more even within the same establishment, a premium that reflects the large share of men among CBW. Once we condition on gender in column 9, this gap disappears.

Second, we examine the level of wages among a cross-section of Swiss prime-age workers in the SESS to understand whether the reform depressed wages more broadly. While native's wages are a central object of interest when studying immigration effects, they naturally conflate potential impacts from increased competition through immigrants and potential selection effects among those who remain in the workforce. This requires particular caution when interpreting impacts on female

workers, a group that experiences sharp compositional changes following the reform (see Appendix Figure C3 and our main results). Appendix Figure D7 estimates reform impacts on log hourly wages for male and female workers. In Panel (a), we observe no impact on male wages and an imprecisely estimated impact of -2.2 log points for women who remain in the workforce. In Panels (b) and (c), we split the sample by whether workers are on an individually negotiated contract, for which wage adjustments could be more easily realized, and workers on collectively bargained contracts that do not allow for individual flexibility and limit frequent compensation adjustments. We see positive point estimates for male and female workers on individually negotiated contracts, who comprise about half of the sample (Panel b). Instead, the negative point estimates are driven by the group whose contracts limit firms' leeway in setting wages. This provides suggestive evidence that wage impacts in the overall sample may rather be the result of compositional changes among female workers. Even if we were take the estimate of -2.2 log points for female workers in Panel (a) at face value, the implied labor supply elasticity at the extensive margin is around 4 (9% reduction in workforce participation over a 2.2% reduction in wages). This value is an order of magnitude above extensive margin elasticity estimates for married women which tend to range between .2 – .6 in industrialized countries (Bargain, Orsini, and Peichl, 2014). Taken together, wage pressure alone is therefore unlikely to explain maternal drop-out in this context.

5.5.2 Household Level Income Shocks

Could income effects within households explain mothers' dropout? As shown by prior work studying the impact of this reform on the firm side (Beerli et al., 2021), highly educated native workers in the border region benefited from wage growth. This raises the possibility that mothers' household income is increasing after the reform, thus making it financially less attractive or necessary to participate in the labor market.

To examine the importance of potential household income effects, we estimate the reform impact on partner's earnings for mothers in our main sample. To do so, we match mothers to their partners at baseline and restrict to women whose partner is Swiss. Appendix Figure D8 documents that the partners of these mothers experience an increase in yearly wage earnings of about 1.4% in the transition and 1% in the free period. However, these income effects are relatively small compared to the extent of mother's dropout: If we take these estimates at face value, the 1% increase in partner's income in 2010 would need to fully account for maternal drop-out of -5.2 ppt.

To gauge whether such magnitudes might be plausible, we examine the semi-elasticity of mother's workforce participation with respect to partner's earnings in Appendix Table D7. We start by examining this relationship in the cross-section, using our full sample of mothers and their partners in the pre-reform period in Column 1. A 1% increase in partner's earnings is associated with a decrease in maternal participation by .02 ppt. This estimate is two orders of magnitudes smaller than what would be needed to explain our main participation effect of -5.2 ppt. Since a concern in the cross-section might be that this reflects selection, in Column 2 we move to a

panel by adding individual level fixed effects. This leaves the semi-elasticity essentially unchanged (–.016 ppt). We obtain a similar estimate (–.014 ppt) when restricting to the control region in the pre-treatment period in Column 3, and a somewhat larger estimate (–.027 ppt) when using data for the full time period of our panel (1994 - 2010) for the control group. While the partners of the mothers in our setting thus experience moderate earnings growth post-reform, we deem it unlikely that mother's dropout would be driven purely by this channel given the magnitude of these semi-elasticities.

5.5.3 Fertility Impacts

We further shed light on whether differential fertility may matter for the interpretation of our main effect. To do so, we define an indicator variable equal to one if a mother gives birth in a particular year post-reform and zero otherwise. As depicted in Appendix Figure D9, we observe an increase in the likelihood of giving birth for treated mothers in any given year, especially in the *Free* period. If increased fertility following the reform is the main driver behind mothers' decline in participation, we would expect our labor supply effects to be driven by those mothers who have additional children.

We therefore examine heterogeneous treatment effects by splitting our sample into mothers who are at or below median age (37 and younger) and mothers who are above median age at baseline. Younger mothers are substantially more likely to have additional children post-reform: Their likelihood of a post-reform birth is at 17%, compared to 5% for mothers who are above median age. Indeed, as documented in column 2 and 3 in Appendix Table D8, the differential fertility in the treated region can be entirely attributed to younger mothers. If increased fertility were to be the primary reason for mothers' dropout, we would thus expect younger mothers to drive this result. However, when we examine workforce participation by mother's age in columns 4 and 5, we observe drop-out both for older and younger mothers. We therefore deem it unlikely that increased fertility *per se* fully explains our main treatment effects.

6 Effects on the Child Penalty

Lastly, we relate our main estimates to recent work on the so-called child penalty (Kleven, Landais, and Søgaard (2019)) and document that the reform translates into an increase in the penalty in labor force participation for women in treated local labor markets.[14] To do so, we construct a sample of mothers with a first birth between the years 1989 and 2000. We restrict our sample to mothers who at the time of their first birth are between 25 and 40 years old, and follow their labor market participation starting 5 years before and 10 years after their first birth.

[14] We code individuals as in the labor force if they have positive earnings in a spell of wage employment (including jobs that are not obligated to pay into social security), self employment, receive payments because of military duty, maternity leave or unemployment.

Our goal is to estimate whether the penalty in participation for women who live in treated municipalities is changing in response to the reform. Depending on the birth cohort of their child, the reform's impact will materialize at different event times after childbirth. To illustrate the estimation strategy, Figure E1 shows the participation penalties for mothers with a first birth in 1992 and 1996, with the grey dashed lines indicating the timing of the reform for children of these two birth cohorts.[15] For mothers with a first birth in 1992, the last pre-reform year (1999) corresponds to $t+7$ after the birth, while for mothers with a first birth in 1996, the last pre-reform year is $t+4$. The penalty in participation is generally larger for mothers in control municipalities. However, the relative penalty between treated and control regions starts to shrink once the reform takes effect. For mothers with a first birth in 1996, the child penalty between treated and control eventually reverses at $t+10$ after the birth.

We then estimate a triple difference in the participation penalty between treatment and control mothers: For each event time t, we estimate how the child penalty changes when that event time happens post- rather than pre-reform for treated vs. control mothers.[16] Following Thakral and Tô (2026), we estimate child penalties at event time t while controlling flexibly for mother's concurrent age, mother's age at birth, and the child's birth cohort via imposing an additional linear restriction: We assume no anticipation effects of childbirth at $t-5$. We thus estimate the following equation, with y_{imt} an indicator for labor force participation for mother i at event time t, who lives in municipality m at baseline, and whose first child belongs to birth cohort c:

$$y_{imt} = \sum_{j=0}^{10} \alpha_j \cdot Post_{cj} \cdot Treat_m \cdot \mathrm{I}\left[\mathrm{j} = \mathrm{t}\right] + \sum_{\substack{j=-10 \\ j\neq -5 \\ j\neq -1}}^{10} \beta_j \cdot Treat_m \cdot \mathrm{I}\left[\mathrm{j} = \mathrm{t}\right] + \sum_{k\neq 1989} \delta_k \cdot Treat_m \cdot \mathrm{I}\left[\mathrm{k} = \mathrm{c}\right]$$
$$+ \sum_{k\neq 1989} \sum_{\substack{j=-10 \\ j\neq -5 \\ j\neq -1}}^{10} \gamma_{kj} \cdot \mathrm{I}\left[\mathrm{j} = \mathrm{t}\right] \cdot \mathrm{I}\left[\mathrm{k} = \mathrm{c}\right] + \sum_{k\neq 1989} \zeta_k \cdot \mathrm{I}\left[\mathrm{k} = \mathrm{c}\right] + \eta\, Treat_m + \psi_i + \phi_{it} + \epsilon_{imt} \tag{3}$$

$Post_{ct}$ is an indicator that is equal to one if a child's birth cohort's event time t occurs post-reform, $Treat_m$ is an indicator equal to one if the mother lives in a treated municipality at baseline, $\mathrm{I}\left[\mathrm{j} = \mathrm{t}\right]$ are event time indicators and $\mathrm{I}\left[\mathrm{k} = \mathrm{c}\right]$ are (child) birth cohort indicators. We control for mother's age at birth with ψ_i and mother's age in year t with ϕ_{it}. The coefficients of interest, α_j, measure the change in the child penalty at each event time t for mothers in border municipalities post-reform. Result are reported in Appendix Table E1. Columns 1 and 2 report results when we use the full sample, while Columns 3–4 reports results when we drop data from the year 1998 (including the corresponding birth cohort for whom this is the baseline year (birth cohort 1999), see Section 3).

[15] Following Thakral and Tô (2026), we control flexibly for mother's concurrent age and mother's age at birth via imposing no anticipation effects of childbirth at $t-5$.

[16] We do not have variation in treatment status and can therefore not estimate treatment effects in the post period for event times $t=-5$ to $t=-1$, as these refer to years in which all birth cohorts in our sample are untreated.

In columns 2 and 4, we estimate the "traditional" child penalty by assuming no impact of mother's age and estimating $t-5$. Qualitatively, results across specifications and samples are very similar: The participation penalty for treated mothers increases by between 1-2 ppt when this event time occurs in the post-reform period. These effects are somewhat larger and more precisely estimated for event times closer to the birth (i.e. in the first 5 years) and sizable: One year (five years) after a mother's first birth, this corresponds to an increase of about 8.8% (6.5%) relative to the average child penalty in the control group (-.29). Consistent with our main results, the child penalty for mothers in border municipalities meaningfully *increases* following the reform.

7 Conclusion

In this paper, we use a large and geographically concentrated expansion in the supply of full-time cross-border workers to trace out firms' demand for part-time labor. We show that Swiss mothers, who work predominantly part-time, drop out of the workforce, while other prime-age groups, who work predominantly full-time, are unaffected. Several pieces of evidence indicate that mothers drop out and stay out because firms reduce their demand for part-time workers once full-time labor becomes more readily available. Reduced hours thus appear to be an amenity that is on average costly for firms to provide, with mothers bearing the cost: the participation child penalty rises by around 6% as a result of the reform.

Our findings add a firm-side perspective to work on policies that regulate working time, some of which are aimed specifically at integrating maternal talent (Ciasullo and Uccioli, 2023; Wasserman, 2023; Carry, 2026). To the extent that firms are not compensated for accommodating shorter or more flexible hours, such policies may carry unintended productivity consequences. Our results also complement work that approaches job amenities from the worker side (Mas and Pallais, 2017; Wiswall and Zafar, 2018; Maestas et al., 2023; Sorkin, 2018). That literature estimates what workers would pay for shorter or more flexible hours while holding the supply of such jobs fixed. We show that this supply is itself responsive: when full-time labor becomes available, firms restructure the types of position offered.

While our study implies that reduced hours jobs can be costly for firms to provide, an important path for future research lies in documenting the magnitude of such costs. From a policy perspective, this would help to better assess the trade-offs associated with the provision of low(er) hours jobs. More generally, our results are consistent with the view that how firms package tasks into jobs is a choice rather than a fixed feature of production.

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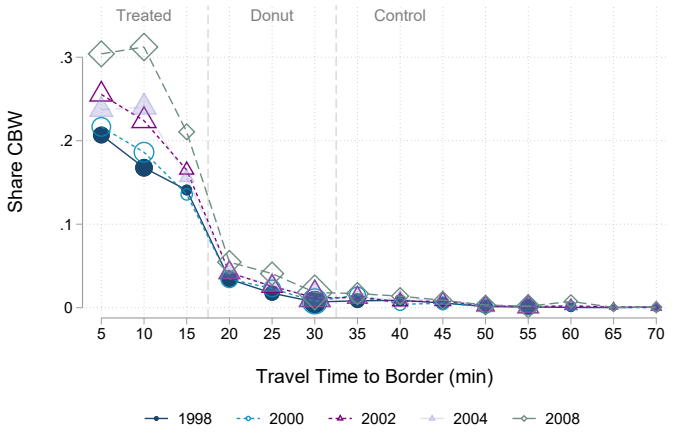
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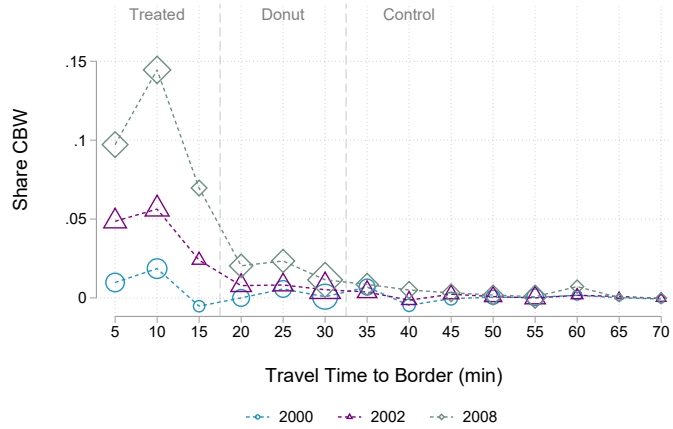
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Tables and Figures

Figure 1: Municipal Share CBW by Travel Time to Border



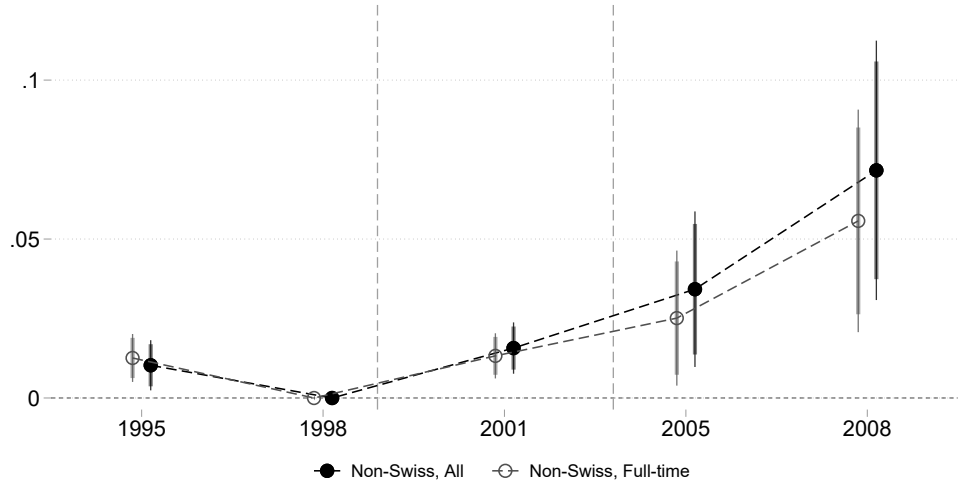
(a) Share CBW by Year



(b) Increase in CBW from 1998

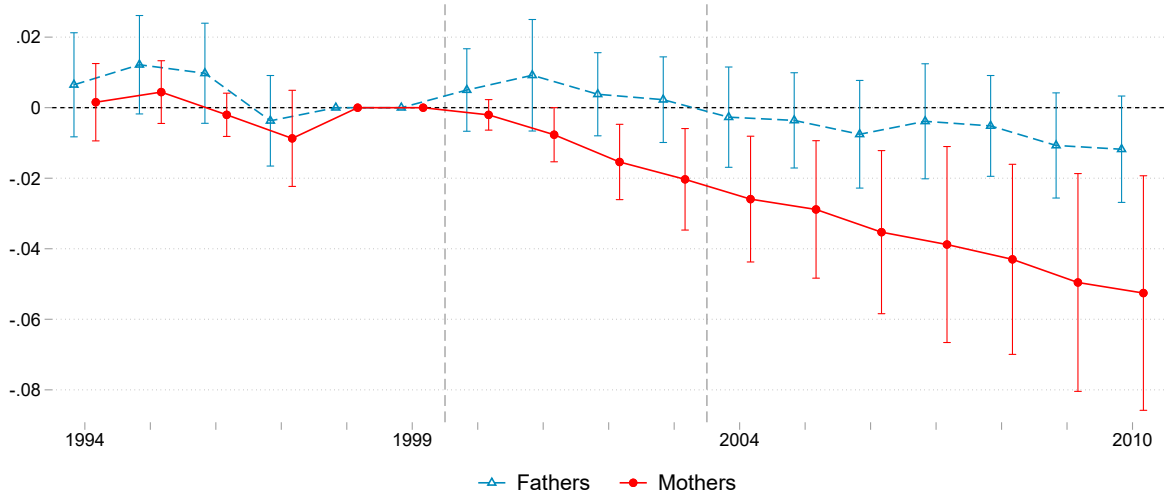
Notes: This figure shows the share of cross-border workers (CBW) and the increase in CBW by travel time by car to the nearest border crossing in minutes. The share CBW is the total number of CBW in a municipality and year divided by total workers in that municipality in 1998. Panel (a): Share CBW in levels. Panel (b): Change in the share CBW relative to the share CBW in 1998. Each travel distance bin is weighted by total workers in 1998. Dashed vertical lines indicate Treated ($\leq$ 15 min), Donut ($>$ 15-30 min) and Control ($>$ 30 min) municipalities. Based on SESS data.

Figure 2: Reform Impact on Share of Non-Swiss Workers in Local Labor Markets



Notes: Estimates of Equation 1. This figure shows the reform's impact on the share of non-Swiss workers and non-Swiss full-time workers. The outcome is the total number of workers in a specific category in a municipality and year divided by total workers in that municipality in 1998. Each municipality is weighted by number of workers in 1998. All specifications include municipality and year fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Business Census data.

Figure 3: Reform Impact on Parental Participation in the Workforce



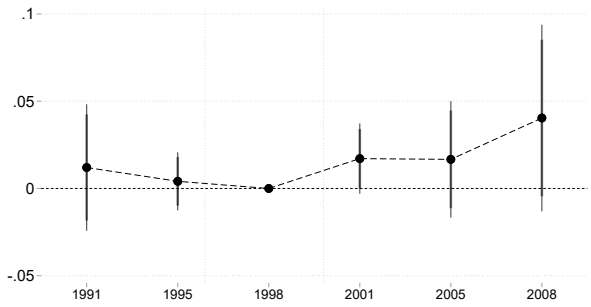
Notes: Estimates of Equation 1 for mothers and fathers (prime-age women and men with a child younger than 16 in 2000). The dependent variable is an indicator for participation in the workforce in a given year, defined as having yearly real earnings above 3,900 CHF (see Section 3.1). All specifications include municipality and year fixed effects, and individual-level controls measured in the year 2000: age, age squared, indicators for education levels (compulsory, secondary, tertiary), bins for total number of children and bins for age of the youngest child. Standard errors clustered at the CZ level and 95% confidence intervals displayed. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Social Security data.

Table 1: Reform Impact on Participation in the Workforce: Difference-in-Differences

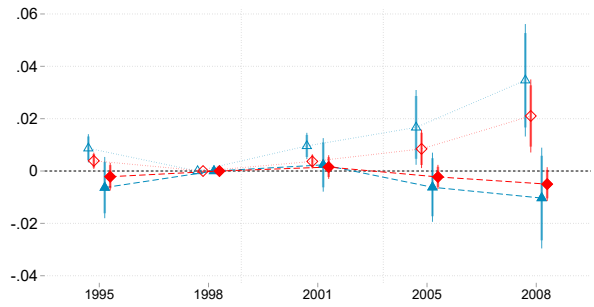
	Mothers						Fathers					
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Free * Treat	–0.038***	–0.038***	–0.038***	–0.030***	–0.032***	–0.032***	–0.011***	–0.011***	–0.011***	–0.007**	–0.011***	–0.011***
	(0.012)	(0.012)	(0.012)	(0.008)	(0.010)	(0.011)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Transition *	–0.010*	–0.010*	–0.010*	–0.006	–0.012**	–0.011*	0.000	0.000	0.000	0.002	–0.000	–0.001
Treat	(0.005)	(0.005)	(0.005)	(0.004)	(0.006)	(0.005)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Individ. Controls		X		X	X	X		X		X	X	X
Individual FE			X						X			
Nuts-II Trend				X						X		
LFP Trend					X						X	
No city						X						X
Adj. R^2	0.036	0.069	0.502	0.037	0.037	0.069	0.016	0.027	0.425	0.016	0.016	0.023
Obs	8,018,192	8,018,192	8,018,192	8,018,192	8,018,192	7,584,528	7,224,800	7,224,800	7,224,800	7,224,800	7,224,784	6,860,112
Dep mean	0.583	0.583	0.583	0.583	0.583	0.579	0.965	0.965	0.965	0.965	0.965	0.967

Notes: Estimates of Equation 2 for mothers and fathers. Outcome is an indicator for participation in the workforce. Transition is an indicator for the transition period (2000-2003), Free is an indicator for the fully liberalized period (2004-2010). Columns 1 and 6: Municipality and year fixed effects only. Columns 2 and 7: Individual level controls measured at baseline. Columns 3 and 8: Individual level fixed effects. Columns 4 and 9: Nuts-II time trends. Columns 5 and 10: Baseline maternal workforce participation at the municipal level interacted with year fixed effects. Columns 6 and 12: Excluding the three largest cities at baseline (Zurich, Geneva, Basel). Dep mean is the pre-reform mean of the dependent variable. All specifications contain municipality and year fixed effects, and individual level controls measured at baseline unless otherwise noted above. Standard errors clustered at the CZ level. Based on Social Security data. * p<0.1, ** p<0.05, *** p<0.01.

Figure 4: Reform Impact on Job Shares in Local Labor Markets



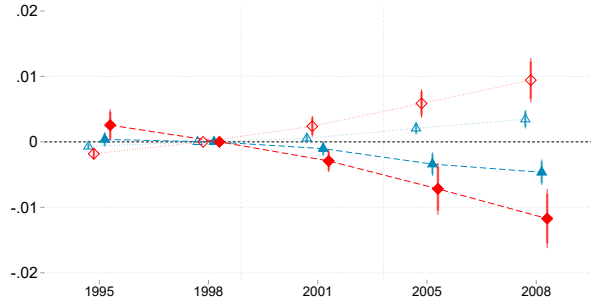
Panel 1a: Full-time ($\geq$ 90% of a FTE)



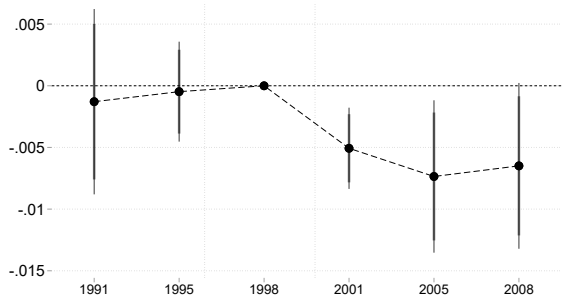
Panel 1b: Full-time by Gender



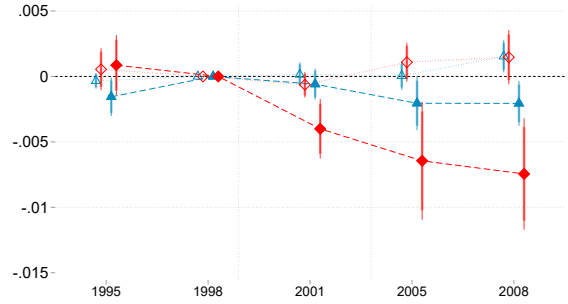
Panel 2a: Part-time I ($\geq$ 50% of a FTE)



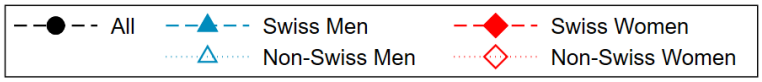
Panel 2b: Part-time I by Gender



Panel 3a: Part-time II ($<$ 50% of a FTE)

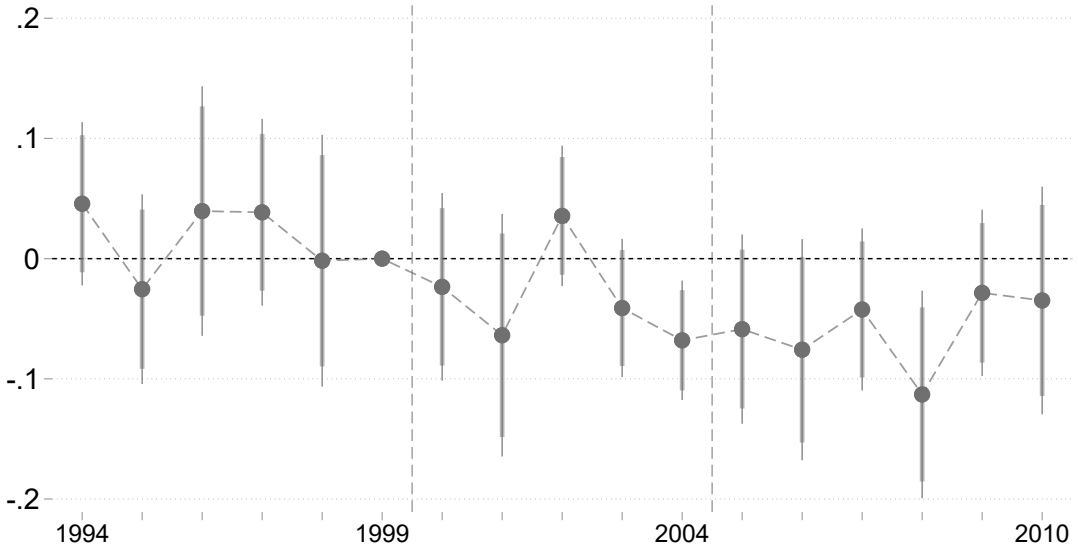


Panel 3b: Part-time II by Gender



Notes: Estimates of Equation 1. Outcome is the number of workers in a specific category in a municipality and year over total workers in 1998 in that municipality. Panel 1: Full-time workers (employment level of $\geq$ 90% of an FTE). Panel 2: Part-time I workers (employment level between 50% - 89% of an FTE). Panel 3: Part-time II workers (employment level $<$ 50% of an FTE). Each municipal observation is weighted by total number of workers in 1998. All specifications include municipality and year fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Business Census data.

Figure 5: Reform Impact on Labor Demand: Job Advertisement for Part-time Position



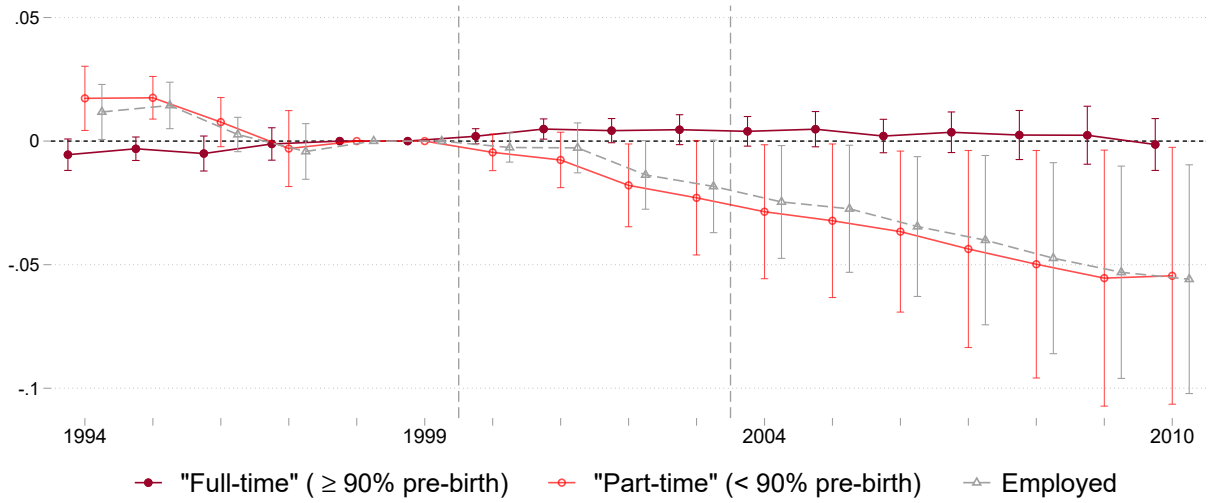
Notes: Estimates of Equation 1. Outcome is an indicator for whether a job advertisement is for a part-time position (employment level < 90% of an FTE). The specification includes year, major region and industry fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on SJMM data (Buchmann et al., 2022*a*).

Table 2: Reform Impact on Labor Demand: Type of Position Advertised

	Full-time	Part-time I ($\geq$ 50% FTE)	Part-time II (< 50% FTE)	Empl. Level % of FTE	Managment Position	SIOPS Score	Mid/High Prestige
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Free * Treat	0.082**	–0.047*	–0.035**	2.926**	0.027	1.607**	0.066**
	(0.034)	(0.025)	(0.016)	(1.458)	(0.017)	(0.807)	(0.030)
Transition * Treat	0.047	–0.021	–0.026	1.995	0.007	–0.217	–0.010
	(0.038)	(0.023)	(0.019)	(1.794)	(0.024)	(0.805)	(0.030)
Adj. R^2	0.146	0.117	0.035	0.085	0.019	0.314	0.176
Obs	20,040	20,040	20,040	20,040	20,040	20,263	20,263
Dep mean	.878	.092	.03	94.901	.108	40.36	.164

Notes: Estimates of Equation 1. Outcomes are: Indicators for whether a job is advertised as full-time (column 1), high part-time (column 2) or low part-time (column 3), employment level in % of an FTE (column 4), indicator for a position with supervisory or leadership functions (column 5), Standard International Occupational Prestige Scale (SIOPS, see Montgomery (1988)) of the advertised position (column 6), indicator for upper-middle to high prestige (SIOPS score $\geq$ 51). Dep mean is the pre-reform period mean of the dependent variable. All specifications include year, major region and industry fixed effects. Standard errors clustered at the CZ level. Based on SJMM data (Buchmann et al., 2022*a*). * p<0.1, ** p<0.05, *** p<0.01.

Figure 6: Reform Impact on the Composition of Employed Mothers



Notes: Estimates of Equation 1. The sample consists of wage employed mothers with a first birth after 1982 in post-birth years. Mothers are classified as working full-time vs part-time based on the "post-birth earnings share", i.e. the share of pre-birth earnings recovered upon returning to employment after childbirth (see Section 5.3). All specifications include municipal and year fixed effects and individual level controls measured at baseline. Standard errors clustered at the CZ level and 95% confidence intervals displayed. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on social security data.

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A Descriptives

Table A1: Summary Statistics: Prime-Age Adults

	Non-Fathers	Fathers			Non-Mothers	Mothers		
			Border	Central			Border	Central
Demographics								
Census								
Age	36.26	40.23	40.38	40.18	36.83	38.09	38.34	37.98
	(8.21)	(6.21)	(6.23)	(6.19)	(8.61)	(5.92)	(5.96)	(5.89)
Age at First Birth	.	30.19	30.34	30.11	.	27.74	27.99	27.59
	(.)	(4.72)	(4.83)	(4.65)	(.)	(4.44)	(4.56)	(4.36)
Number of Children	0.00	2.12	2.07	2.15	0.00	2.11	2.06	2.16
	(0.00)	(0.85)	(0.84)	(0.86)	(0.00)	(0.85)	(0.83)	(0.85)
Age Youngest Child	.	6.58	6.55	6.59	.	6.87	6.89	6.85
	(.)	(4.67)	(4.63)	(4.68)	(.)	(4.67)	(4.63)	(4.68)
Obligatory Schooling	0.10	0.07	0.07	0.07	0.11	0.13	0.12	0.14
	(0.29)	(0.26)	(0.25)	(0.26)	(0.32)	(0.34)	(0.33)	(0.34)
Secondary Degree	0.53	0.52	0.51	0.53	0.57	0.64	0.61	0.65
	(0.50)	(0.50)	(0.50)	(0.50)	(0.50)	(0.48)	(0.49)	(0.48)
Tertiary Degree	0.32	0.40	0.41	0.39	0.27	0.21	0.25	0.19
	(0.47)	(0.49)	(0.49)	(0.49)	(0.44)	(0.41)	(0.43)	(0.39)
Labor Market								
Social Security Data								
In Workforce 1999	0.91	0.96	0.95	0.97	0.88	0.60	0.61	0.59
	(0.29)	(0.18)	(0.23)	(0.17)	(0.33)	(0.49)	(0.49)	(0.49)
Any Unemployment Benefits 1999	0.05	0.03	0.04	0.03	0.05	0.03	0.04	0.02
	(0.22)	(0.17)	(0.19)	(0.16)	(0.22)	(0.18)	(0.20)	(0.15)
Median Earnings (CHF) | Employed 1999	63,141	79,001	81,218	77,000	53,870	24,763	29,294	22,820
	(35,868)	(41,915)	(43,726)	(41,239)	(27,932)	(24,118)	(27,472)	(22,404)
Census								
Works Part-Time	0.09	0.06	0.06	0.06	0.30	0.81	0.77	0.82
	(0.29)	(0.23)	(0.23)	(0.23)	(0.46)	(0.39)	(0.42)	(0.38)
Weekly Hours	43.58	46.17	45.10	46.75	38.37	22.69	24.19	22.10
	(9.81)	(10.40)	(9.56)	(10.98)	(10.94)	(16.05)	(15.08)	(16.73)
Commute time in minutes	22.16	20.21	20.18	19.82	23.64	16.07	16.67	15.43
	(18.53)	(18.20)	(17.35)	(18.42)	(17.87)	(15.15)	(14.51)	(15.26)
Commutes to other area group	0.04	0.04	0.02	0.01	0.04	0.03	0.01	0.00
	(0.20)	(0.20)	(0.15)	(0.09)	(0.20)	(0.17)	(0.10)	(0.06)
Industry								
Agriculture, Mining, Energy	0.05	0.08	0.06	0.09	0.02	0.04	0.03	0.05
	(0.21)	(0.27)	(0.24)	(0.29)	(0.12)	(0.20)	(0.18)	(0.22)
Construction	0.08	0.08	0.07	0.09	0.01	0.02	0.02	0.02
	(0.27)	(0.28)	(0.25)	(0.29)	(0.12)	(0.15)	(0.14)	(0.15)
Manufacturing	0.18	0.19	0.18	0.20	0.10	0.07	0.08	0.07
	(0.39)	(0.39)	(0.38)	(0.40)	(0.30)	(0.26)	(0.27)	(0.26)
Non-Knowledge-Int. Services	0.20	0.20	0.19	0.20	0.23	0.23	0.21	0.23
	(0.40)	(0.40)	(0.39)	(0.40)	(0.42)	(0.42)	(0.41)	(0.42)
Knowledge-Int. Services	0.24	0.22	0.25	0.20	0.24	0.17	0.19	0.15
	(0.43)	(0.42)	(0.44)	(0.40)	(0.43)	(0.37)	(0.39)	(0.36)
Public/Edu/Health	0.12	0.16	0.17	0.15	0.29	0.31	0.33	0.30
	(0.33)	(0.36)	(0.38)	(0.36)	(0.45)	(0.46)	(0.47)	(0.46)
Missing Industry	0.12	0.07	0.07	0.07	0.11	0.16	0.14	0.17
	(0.32)	(0.25)	(0.26)	(0.25)	(0.31)	(0.37)	(0.35)	(0.38)
Observations	461,730	451,550	94,155	229,810	383,134	501,137	107,764	250,262

Notes: This table shows summary statistics of prime-age men and women in 2000 by parental status: Non-parents without children in 2000 or parents of children younger than age 16 in 2000. Standard deviations displayed in round brackets. Variables based on the Census are measured in 2000, variables based on the Social Security data are measured in 1999.

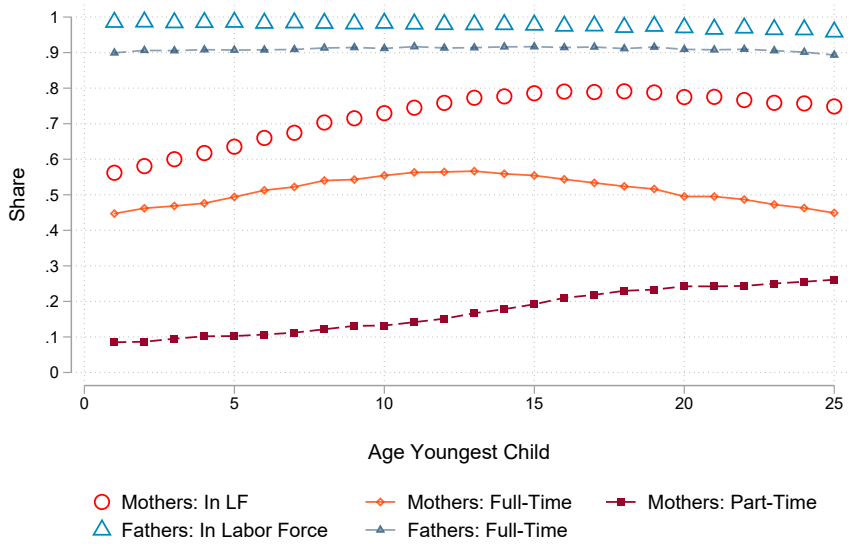
Table A2: Summary Statistics: Establishments and Workers in the Business Census

	Border	Central
A. Establishments, 1998		
Number of Workers (Mean)	9.44	8.43
	(49.83)	(36.87)
Number of Workers (Median)	3.00	3.00
	(.)	(.)
Private Sector Enterprise	0.92	0.90
	(0.28)	(0.30)
Single Establishment	0.79	0.78
	(0.41)	(0.41)
Industry: Construction	0.09	0.10
	(0.28)	(0.31)
Manufacturing	0.11	0.12
	(0.31)	(0.32)
Non-Knowledge-Int. Services	0.41	0.40
	(0.49)	(0.49)
Knowledge-Int. Services	0.26	0.24
	(0.44)	(0.43)
Public/Edu/Health	0.13	0.14
	(0.34)	(0.34)
B. Workers		
Non-Swiss, 1995	0.37	0.18
	(0.11)	(0.08)
Cross-Border Workers, 1995	0.14	0.00
	(0.09)	(0.01)
Non-Swiss, 1998	0.34	0.17
	(0.09)	(0.07)
Full-time ($\geq$ 90% FTE)	0.76	0.73
	(0.06)	(0.06)
Part-time I ($\geq$ 50% FTE)	0.14	0.14
	(0.04)	(0.04)
Part-time II ($<$ 50% FTE)	0.10	0.13
	(0.03)	(0.04)
By Origin, Gender and Employment Level, 1998		
Swiss Men	0.37	0.48
Full-time	0.33	0.43
Part-time I	0.02	0.03
Part-time II	0.02	0.03
Swiss Women	0.28	0.35
Full-time	0.15	0.17
Part-time I	0.08	0.10
Part-time II	0.05	0.09
Non-Swiss Men	0.21	0.10
Full-time	0.20	0.09
Part-time I	0.01	0.00
Part-time II	0.01	0.00
Non-Swiss Women	0.13	0.06
Full-time	0.08	0.04
Part-time I	0.03	0.01
Part-time II	0.02	0.01
Number of Establishments	90,620	170,210
Number of Workers	855,156	1,434,174

Notes: This table shows summary statistics of establishments and workers for the year 1998 unless otherwise indicated. Standard deviations displayed in round brackets. Based on Business Census data.

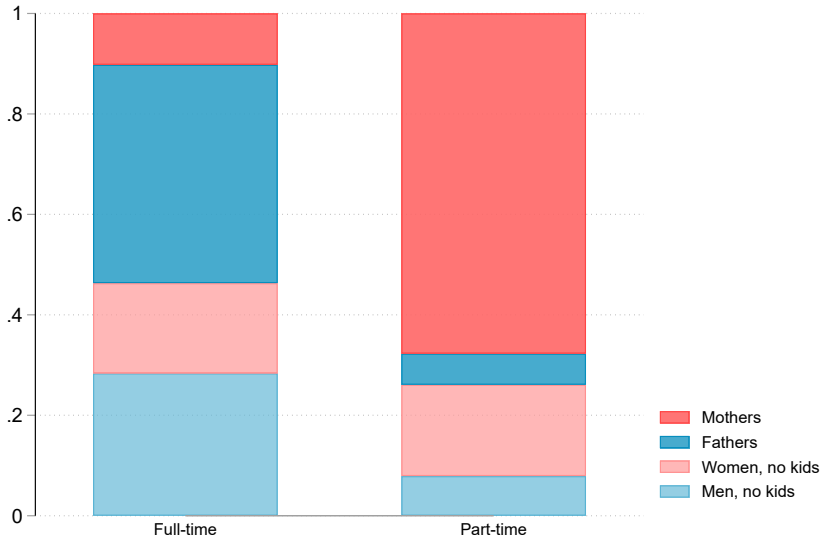
B Setting

Figure B1: Parental Labor Force Participation by Age of Youngest Child



Notes: This figure shows labor force participation rates of prime-age Swiss mothers and fathers by age of their youngest child in the year 2000 based on self-reported information in the Census. Full-time is an indicator equal to one if respondent indicates hours in a full-time job. Part-time is an indicator equal to one if respondent indicates hours in a part-time job and no hours in a full-time job. Respondents that do not indicate hours are set to missing for Full-time and Part-time variables. Based on Census data.

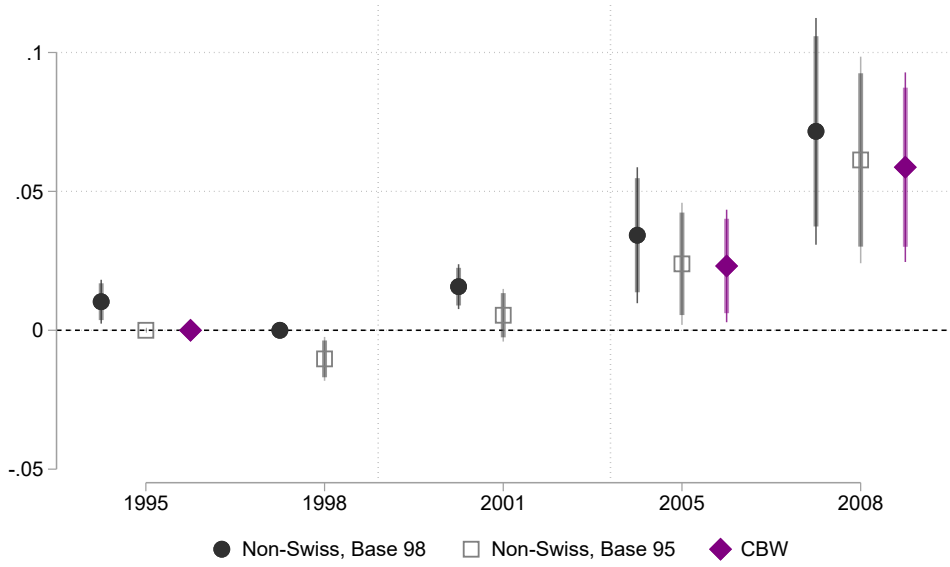
Figure B2: Composition of Prime Age Full- and Part-time Workers



Notes: This figure shows the composition of Swiss prime age full- and part-time workers. A respondent works full-time if they indicate hours in a full-time job and part-time if they indicate hours in a part-time job and no hours in a full-time job. Based on data from the Census 2000.

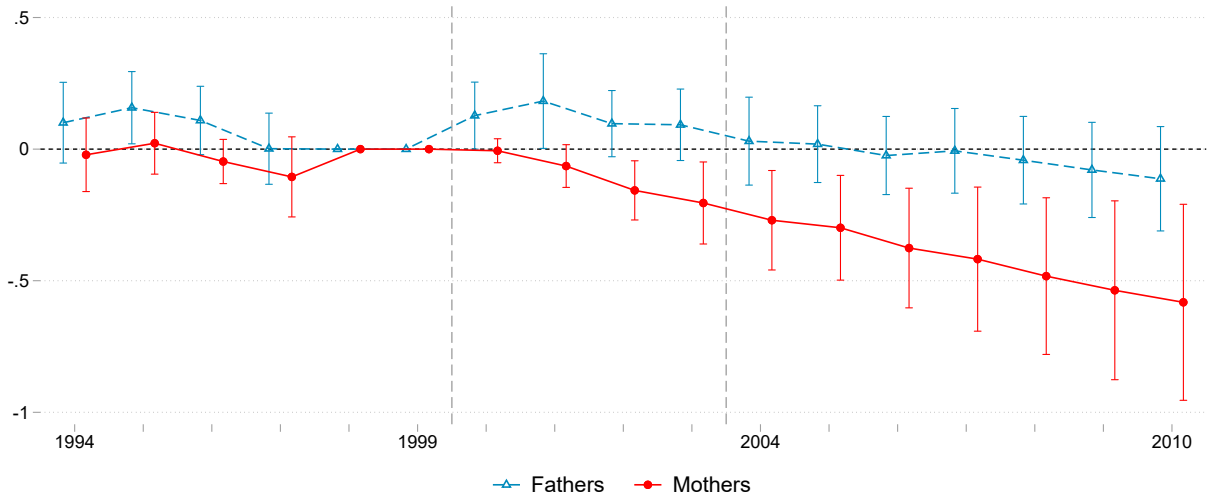
C Main Results: Additional Output

Figure C1: Reform Impact on Non-Swiss Workers and CBW



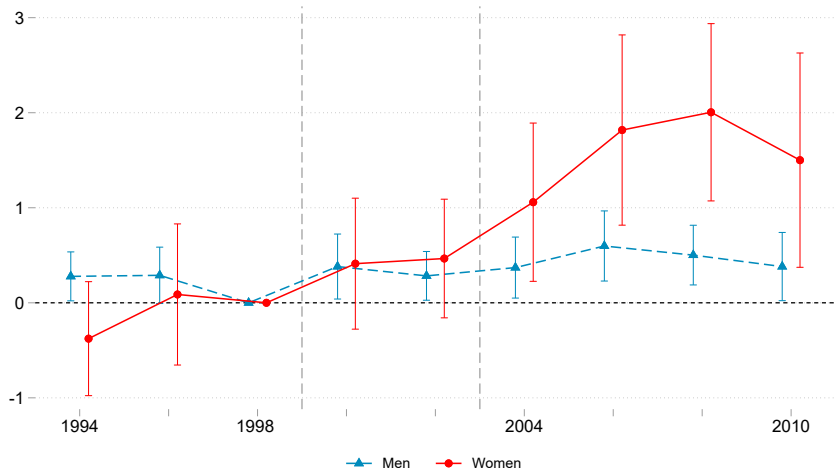
Notes: Estimates of Equation 1. The outcome is the number of non-Swiss or CBW workers in a municipality and year divided by total workers in that municipality in 1998. Information on CBW is only available in 1995, 2005 and 2008 in the Business Census. Estimates are reported with respect to the year 1998 as baseline (Non-Swiss, Base 98; same as displayed in Figure 2) and 1995 (Non-Swiss, Base 95 and CBW). Each municipality is weighted by number of workers in 1998. All specifications include municipality and year fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Business Census data.

Figure C2: Reform Impact on Wage Earnings

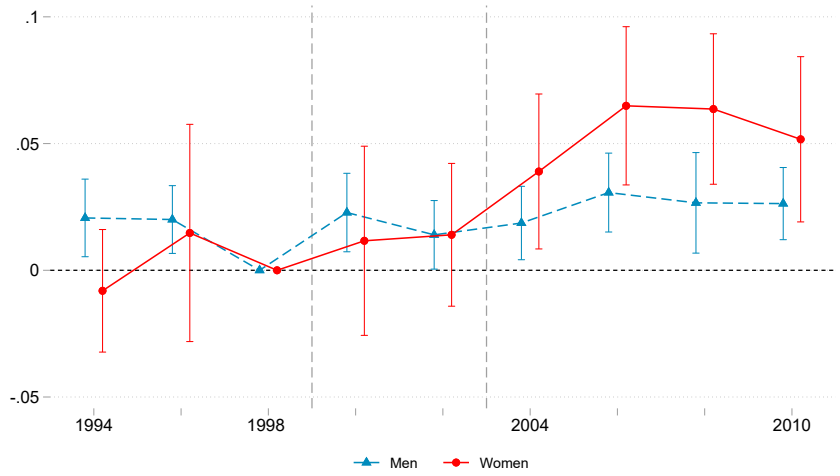


Notes: Estimates of Equation 1 for mothers and fathers. Outcome is the inverse hyperbolic sine of yearly wage earnings across all employment spells. All regressions include municipality and year fixed effects and individual level controls measured at baseline. Standard errors clustered at the CZ level. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on social security data.

Figure C3: Reform Impact on Composition of Employed Women and Men



(a) Weekly Hours



(b) P(Full-time)

Notes: Estimates of Equation 1. The sample includes Swiss workers who are prime-age (25- 54 years old in 2000) and on a regular contracts (i.e., a contract that is not on an hourly basis). Panel (a): Weekly Hours. Panel (b): Indicator for working full-time (employment level $\geq$ 90% of an FTE). All regressions include municipality and year fixed effects and control for age, age squared and level of education (low, mid, high). Standard errors are clustered at the CZ level. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on SESS data.

Figure C4: Reform Impact on Workforce Participation for Prime-Age Non-Parents



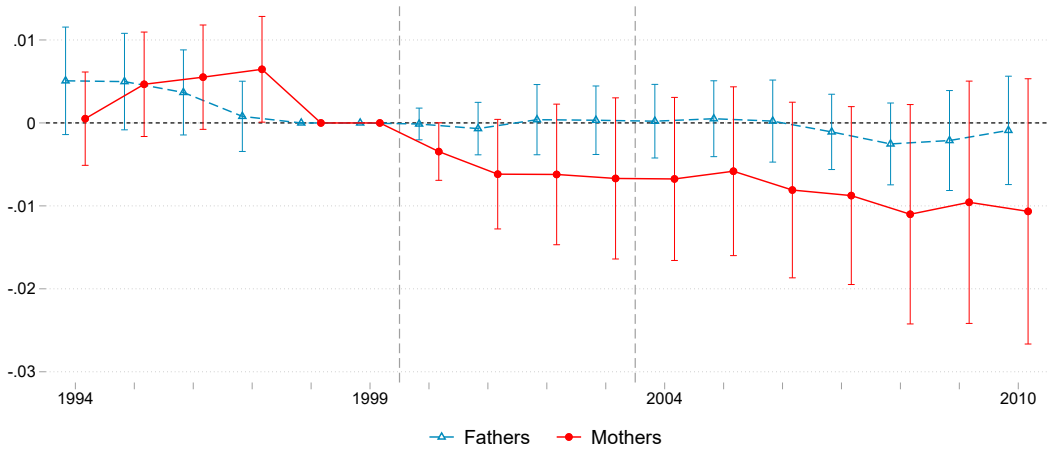
Notes: Estimates of Equation 1 for prime-age men and women who do not have a child at baseline. Outcome is an indicator for workforce participation. All specifications include municipality and year fixed effects, and individual-level controls measured in the year 2000: age, age squared, indicators for education levels (compulsory, secondary, tertiary). Standard errors clustered at the CZ level and 95% confidence intervals displayed. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Social Security data.

Table C1: Robustness: Reform Impact on Workforce Participation, Industry and Occupation Fixed Effects

		Industry		Occupation		Learned Occupation	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Female * Free * Treat	–0.038***	–0.038***	–0.038***	–0.038***	–0.031***	–0.038***	–0.033***
	(0.012)	(0.012)	(0.001)	(0.012)	(0.001)	(0.012)	(0.001)
Female * Transition * Treat	–0.010*	–0.010*	–0.016***	–0.010*	–0.013***	–0.010*	–0.006***
	(0.005)	(0.005)	(0.001)	(0.005)	(0.001)	(0.005)	(0.001)
Male * Free * Treat	–0.011***	–0.011***	–0.008***	–0.011***	–0.008***	–0.011***	–0.009***
	(0.003)	(0.003)	(0.001)	(0.003)	(0.001)	(0.003)	(0.001)
Male * Transition * Treat	0.000	0.000	0.003***	0.000	0.003***	0.000	0.002*
	(0.003)	(0.003)	(0.001)	(0.003)	(0.001)	(0.003)	(0.001)
Without Missing Ind/Occ			X		X		X
Adj. R^2	0.192	0.298	0.122	0.260	0.109	0.197	0.189
Obs	15,242,992	15,242,992	10,919,216	15,242,992	9,676,944	15,242,992	12,509,120
Dep mean: Female	0.583	0.583	0.745	0.583	0.756	0.583	0.609

Notes: Estimates of Equation 2 for mothers and fathers. Outcome is an indicator for participation in the workforce. Transition is an indicator for the transition period (2000-2003), Free is an indicator for the fully liberalized period (2004-2010). Column 1: Main result (see Table 1). Columns 2 and 3: Industry fixed effects (1-digit), including (Column 2) or excluding (Column 3) individuals with missing industry information. Columns 4 and 5: Occupation fixed effects, including (Column 4) or excluding (Column 5) individuals with missing occupation information. Columns 6 and 7: Learned occupation fixed effects, including (Column 6) or excluding (Column 7) individuals with missing learned occupation information. Dep mean is the pre-reform mean of the dependent variable for mothers. All specifications contain municipality and year fixed effects, and individual level controls measured at baseline. Standard errors clustered at the CZ level. Based on Social Security data. * p<0.1, ** p<0.05, *** p<0.01.

Figure C5: Reform Impact on Receiving Unemployment Benefits for Parents

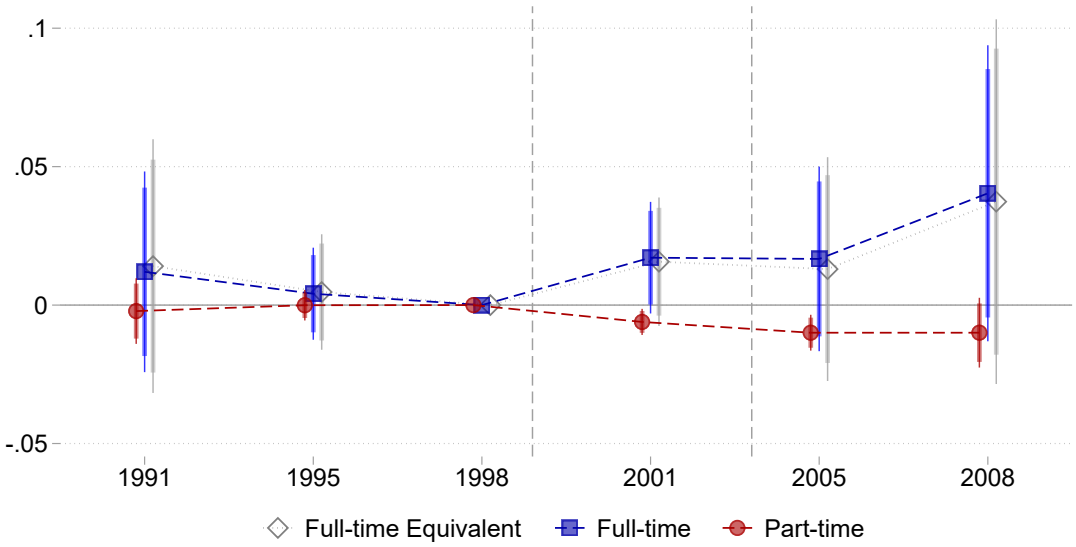


Notes: Estimates of Equation 1 for mothers and fathers. The dependent variable is an indicator for receiving any unemployment benefits in a given year. All specifications include municipality and year fixed effects, and individual-level controls measured at baseline. Standard errors clustered at the CZ level and 95% confidence intervals displayed. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Social Security data.

D Mechanism: Additional Output

D.1 Local Labor Markets

Figure D1: Reform Impact on FTE



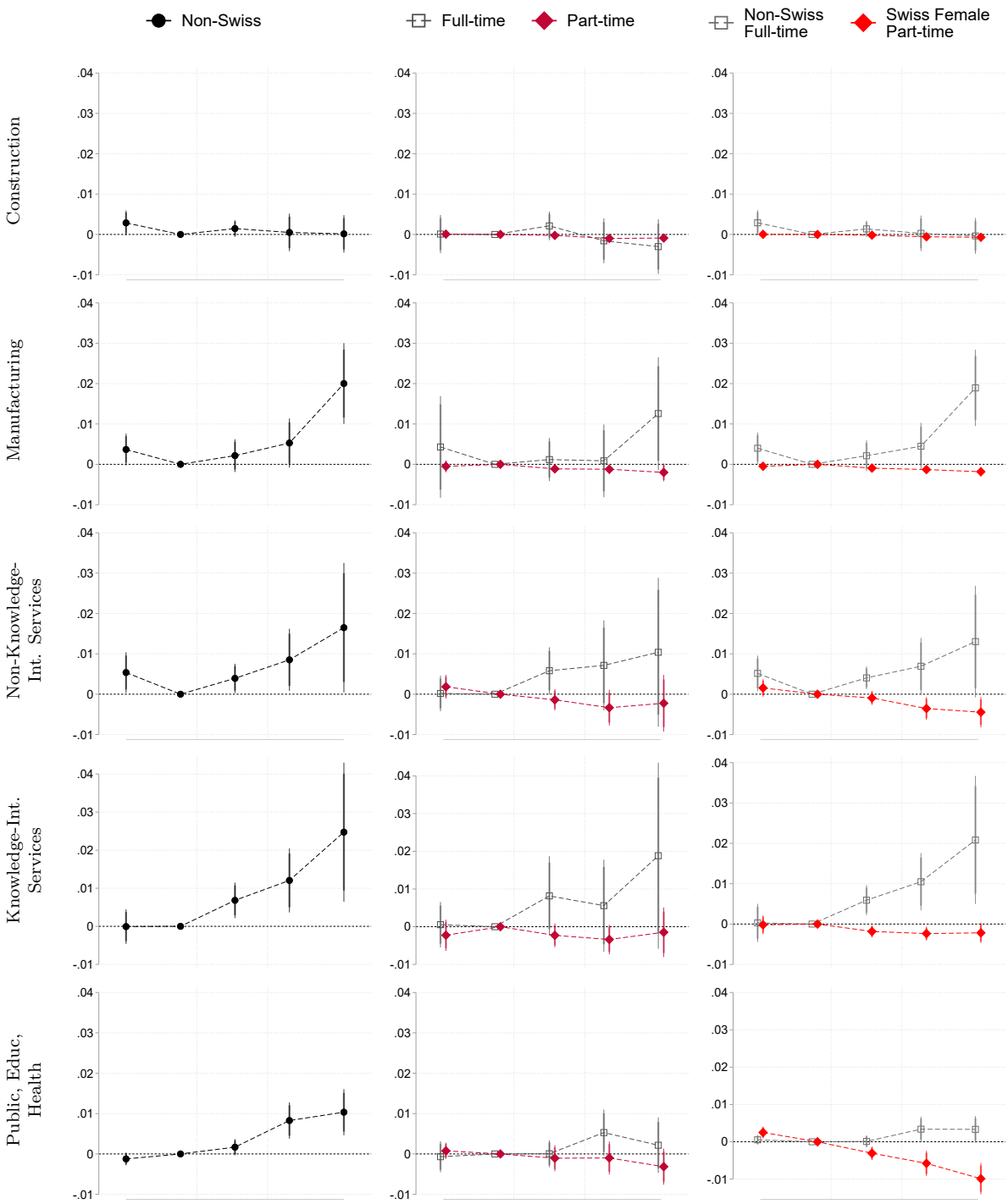
Notes: Estimates of Equation 1. This figure shows the effect of the reform on total hours (FTE) and the share of part- and full-time jobs. Full-time Equivalents (FTE) are the number of FTE in a municipality and year over total FTE in 1998 in that municipality. For Full-time and Part-time, the outcome is the number of workers in a specific category in a municipality and year over total workers in 1998 in that municipality. Each municipal observation is weighted by total number of workers in 1998. All specifications include municipality and year fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Business Census data.

Table D1: Robustness: Local Labor Market Results

Outcome:	Municip. Level (1)	Establ. Level (2)	No Multi-Est. Firms (3)	> 3 Workers (4)	Tight LM (5)	Slack LM (6)
A. Workers						
Full-time	0.026	0.027*	0.032*	0.027*	0.047***	0.002
	(0.025)	(0.015)	(0.018)	(0.015)	(0.016)	(0.016)
Adj. R^2	0.535	0.380	0.395	0.413	0.385	0.378
Dep mean	0.743	0.743	0.744	0.743	0.711	0.758
Part-time	–0.010**	–0.008**	–0.010***	–0.008**	–0.017**	–0.001
	(0.005)	(0.004)	(0.003)	(0.004)	(0.007)	(0.008)
Adj. R^2	0.807	0.428	0.470	0.560	0.471	0.388
Dep mean	0.257	0.257	0.256	0.257	0.289	0.242
Non-Swiss Full-time	0.034**	0.032***	0.038***	0.031***	0.040***	0.014
	(0.013)	(0.008)	(0.009)	(0.008)	(0.007)	(0.011)
Adj. R^2	0.878	0.517	0.528	0.616	0.502	0.529
Dep mean	0.184	0.184	0.200	0.194	0.154	0.203
Swiss Female Part-time	–0.018***	–0.017***	–0.019***	–0.016***	–0.032***	–0.007
	(0.004)	(0.003)	(0.003)	(0.003)	(0.007)	(0.005)
Adj. R^2	0.841	0.430	0.504	0.594	0.529	0.356
Dep mean	0.163	0.163	0.165	0.163	0.189	0.151
B. Hours						
Full-time FTE	0.030	0.030*	0.035*	0.030*	0.054***	0.003
	(0.028)	(0.017)	(0.020)	(0.018)	(0.018)	(0.018)
Adj. R^2	0.391	0.312	0.315	0.318	0.326	0.296
Dep mean	0.860	0.860	0.862	0.860	0.835	0.875
Part-time FTE	–0.006**	–0.006***	–0.007***	–0.006**	–0.016***	0.001
	(0.003)	(0.002)	(0.003)	(0.003)	(0.005)	(0.005)
Adj. R^2	0.858	0.518	0.560	0.628	0.563	0.462
Dep mean	0.140	0.140	0.138	0.140	0.165	0.125
Non-Swiss Full-time FTE	0.039**	0.036***	0.043***	0.035***	0.046***	0.016
	(0.015)	(0.009)	(0.010)	(0.009)	(0.009)	(0.013)
Adj. R^2	0.870	0.494	0.506	0.592	0.483	0.504
Dep mean	0.213	0.213	0.231	0.225	0.181	0.234
Swiss Female Part-time FTE	–0.012***	–0.012***	–0.014***	–0.012***	–0.024***	–0.004
	(0.002)	(0.002)	(0.002)	(0.002)	(0.004)	(0.003)
Adj. R^2	0.864	0.522	0.586	0.651	0.598	0.433
Dep mean	0.089	0.089	0.089	0.089	0.108	0.078
Observations	12,411	1,430,903	1,277,336	691,912	491,638	856,481

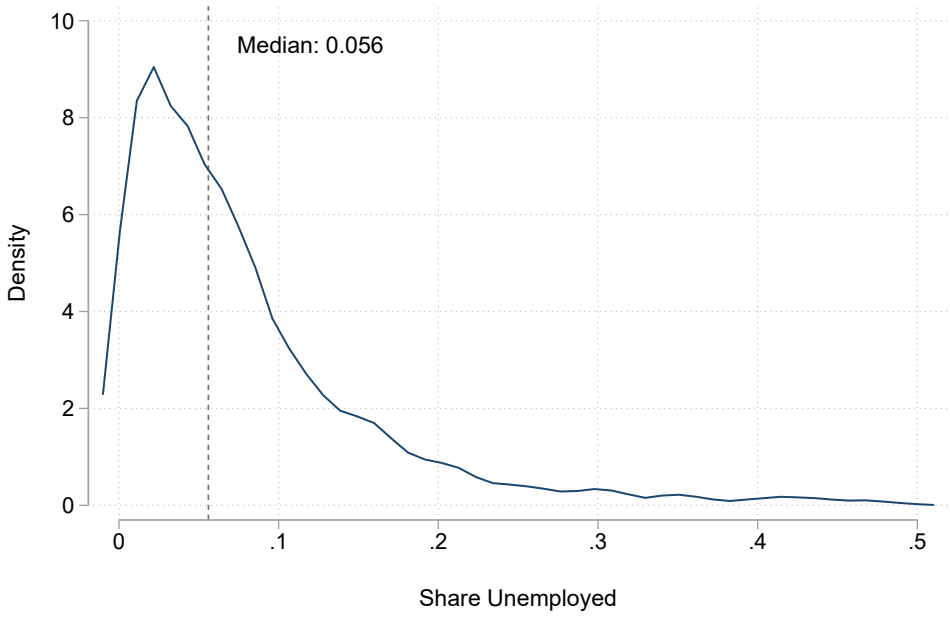
Notes: This table shows the reform impact in the Free period (Equation 2) on workers (Panel A) and hours (Panel B). All outcomes are defined as the number of workers or FTE in a category and year over the total number of workers at baseline. Column 1 reports municipal-level estimates with year and municipality fixed effects, weighted by total workers in a municipality in 1998. Columns 2 – 6 report establishment level estimates with year and establishment fixed effects, weighted by establishment size in 1998. Column 3 drops establishments that belong to firms that operate in both the border and central region. Column 4 restricts to establishments with more than 3 workers at baseline. Columns 5 and 6 restrict to establishments operating in tight vs. slack labor markets pre-reform, respectively. Standard errors are clustered at the commuting zone level. Based on Business Census Data.

Figure D2: Reform Impacts by Industry



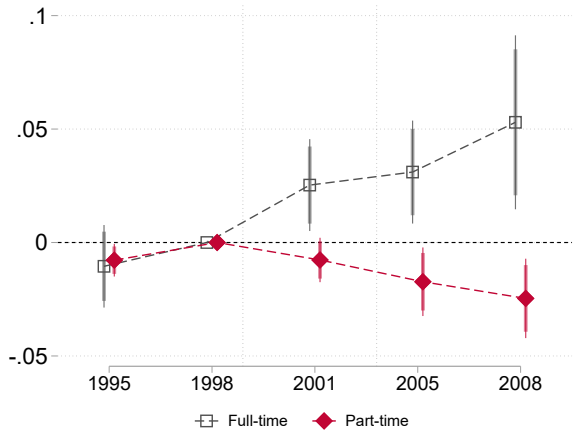
Notes: Estimates of Equation 1 by industry. Outcome is the number of workers in a specific category and industry in a municipality and year over total workers in 1998 in that municipality. Each municipal observation is weighted by total number of workers in 1998. All specifications include municipality and year fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Business Census data.

Figure D3: Tightness: Distribution of Share of Unemployed Workers by Industry-Commuting Zone Cells

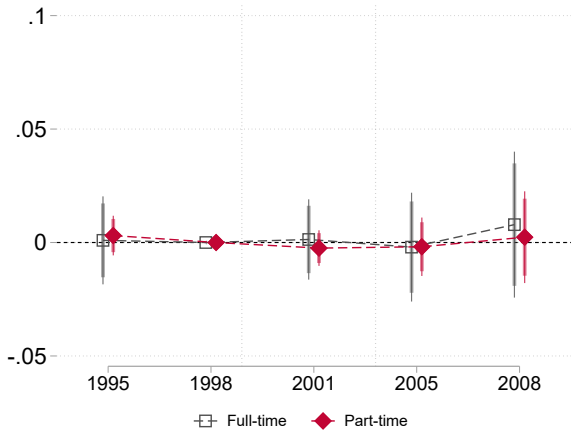


Notes: This figure shows the density of the labor market tightness measure: The average total unemployed mobile workers across 1997 - 1999 within a 3-digit industry-and-commuting-zone cell measured in the unemployment register over the total workers in that cell in 1998 measured in the Business Census. See also Section 5.1.2. Based on Business Census and Unemployment Register data.

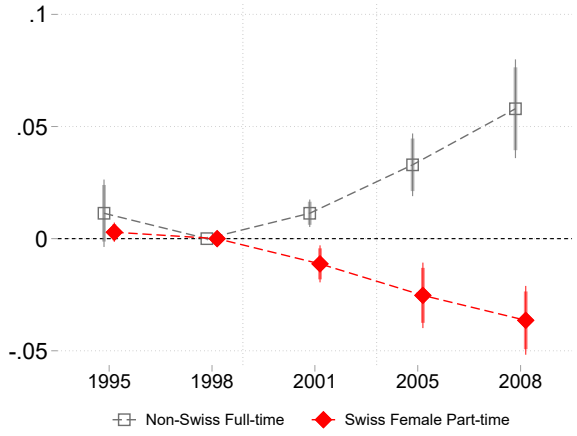
Figure D4: Reform Impact on Workforce Composition by Labor Market Tightness



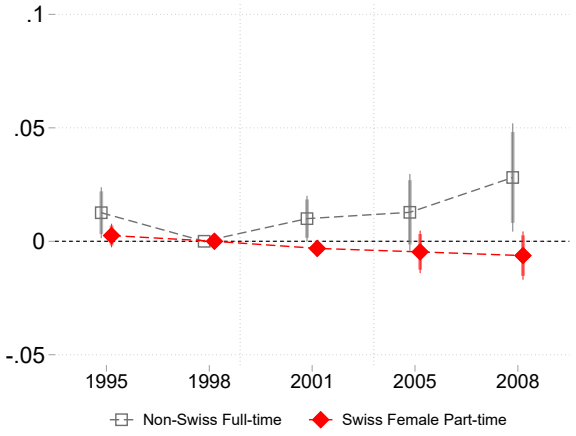
(a) Tight: Full- vs Part-time



(b) Slack: Full- vs Part-time



(c) Tight: Full- vs Part-time by Group



(d) Slack: Full- vs Part-time by Group

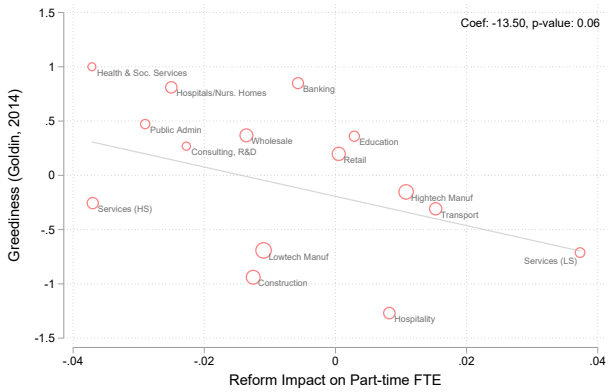
Notes: This figure shows establishment-level reform impacts on different types of workers. Outcome is the number of workers in the specific category and year in an establishment over total workers in 1998 in that establishment. Panels (a) and (c) show results for establishments operating in tight local labor markets pre-reform (above median tightness), Panels (b) and (d) show establishments operating in local labor market with pre-reform slack (at or below median tightness). See Appendix Figure D3 for the construction of the tightness measure. Establishments are weighted by total number of workers in 1998. All specifications include establishment and year fixed effects. Standard errors clustered at the CZ level and bars indicate 90% (thick) and 95% (thin) confidence intervals. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on Business Census and Unemployment Register data.

Table D2: Reform Impacts on Part-time FTE by Industry

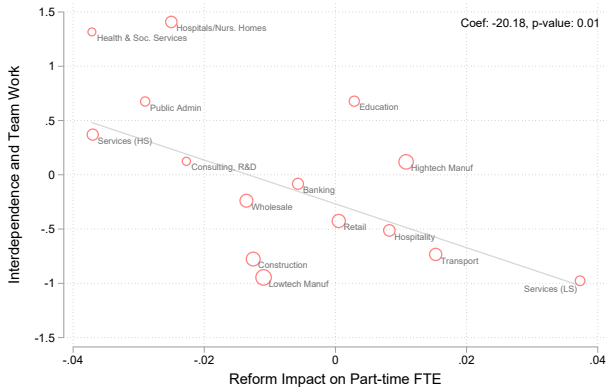
Industry	Share Hours Part-time (1998)	Reform Impact Part-time FTE	SE
Health & Soc. Services	0.38	−0.037***	0.014
Services (HS)	0.13	−0.037***	0.013
Public Admin	0.09	−0.029	0.018
Hospitals/Nurs. Homes	0.32	−0.025*	0.015
Consulting, R&D	0.11	−0.023**	0.010
Wholesale	0.07	−0.014***	0.005
Construction	0.03	−0.013**	0.006
Lowtech Manuf	0.06	−0.011***	0.003
Banking	0.07	−0.006	0.008
Retail	0.21	0.001	0.015
Education	0.38	0.003	0.021
Hospitality	0.09	0.008*	0.004
Hightech Manuf	0.05	0.011**	0.005
Transport	0.10	0.015	0.010
Services (LS)	0.21	0.037*	0.020

Notes: This table reports the reform impact on part-time FTE, i.e. FTE contributed by part-time jobs, across detailed industries. The sample is restricted to establishments operating in tight labor markets pre-reform. Column 1 reports the share of hours that are performed by part-time workers at baseline. Column 2 reports coefficients from a specification based on Equation 2 that estimates separate treatment effects for each industry in the Free period by interacting the *Free* and *Free*Treat* indicators by industry. The outcome is total FTE worked in part-time jobs in an establishment and year over total FTE in that establishment in 1998. Establishments are weighted by the total number of workers in 1998. The specification is run with establishment and year fixed effects. Column 3 reports the standard error of the regression, clustered at the commuting zone level. Based on Business Census Data.

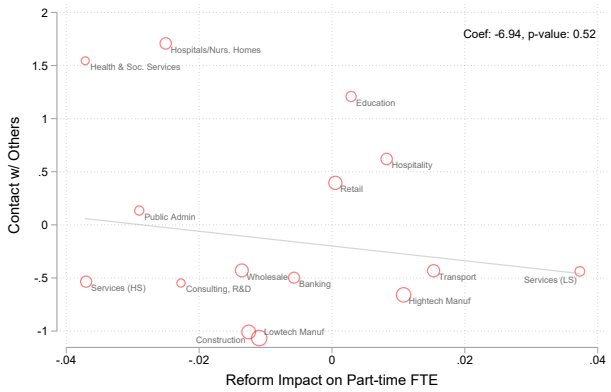
Figure D5: Part-time Reductions and Occupational Characteristics across Industries



(a) "Greediness" based on Goldin (2014)



(b) Interdependence and Team Work



(c) Contact with Others

Notes: This figure shows the relationship between different occupational characteristics (see Section 5.1.3) and the reduction in part-time FTE across industries (see Appendix Table D2). Industries are weighted by total FTE at baseline. Coefficient and robust standard error of a regression of each averaged O*NET measure on the reform impact reported in the upper right corner. Based on Business Census, Census and O*NET data.

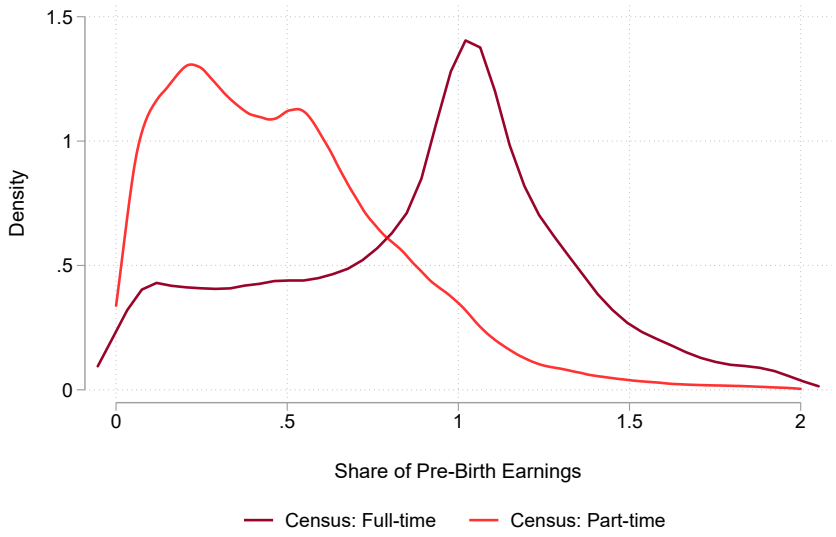
Table D3: Industry Composition

Industry Group	NOGA code (4 digit)	NOGA title	% of Employment
Construction	4531	Installation of electrical wiring and fittings	27.6%
	4533	Plumbing	25.7%
	4521	General construction of buildings and civil engineering works	18.1%
Lowtech Manuf	2030	Manufacture of builders' carpentry and joinery	7.7%
	1551	Operation of dairies and cheese making	5.4%
	2212	Publishing of newspapers	4.2%
Hightech Manuf	2442	Manufacture of pharmaceutical preparations	18.4%
	3350	Manufacture of watches and clocks	16.9%
	3530	Manufacture of aircraft and spacecraft	9.0%
Wholesale	5187	Wholesale of other machinery for use in industry, trade and navigation	11.7%
	5184	Wholesale of computers, computer peripheral equipment and software	10.9%
	5020	Maintenance and repair of motor vehicles	7.4%
Retail	5211	Retail sale in non-specialised stores with food, beverages or tobacco predominating	48.6%
	5212	Other retail sale in non-specialised stores	17.9%
	5248	Other retail sale in specialised stores	8.5%
Hospitality	5510	Hotels	56.4%
	5530	Restaurants	30.2%
	5552	Catering	8.4%
Transport	6411	National post activities	24.6%
	6010	Transport via railways	20.3%
	6420	Telecommunications	16.6%
Services (LS)	7487	Other business activities n.e.c.	24.8%
	7470	Industrial cleaning	18.8%
	9003	Sanitation, remediation and similar activities	11.2%
Banking	6512	Other monetary intermediation	67.4%
	6603	Non-life insurance	24.7%
	6601	Life insurance	3.6%
Consulting, R&D	7412	Accounting, book-keeping and auditing activities; tax consultancy	27.0%
	7414	Business and management consultancy activities	25.2%
	7415	Management activities of holding companies	20.8%
Services (HS)	7222	Other software consultancy and supply	30.9%
	7420	Architectural and engineering activities and related technical consultancy	26.3%
	9131	Activities of religious organizations	12.0%
Health & Soc. Services	8532	Social work activities without accommodation	37.9%
	8514	Other human health activities	24.4%
	8512	Medical practice activities	24.2%
Hospitals/Nurs. Homes	8511	Hospital activities	52.5%
	8531	Social work activities with accommodation	47.5%
Public Admin	7524	Public security, law and order activities	24.6%
	7523	Justice and judicial activities	17.0%
	7522	Defence activities	16.4%
Education	8010	Primary education	38.9%
	8021	General secondary education	22.3%
	8022	Technical and vocational secondary education	13.8%

Notes: This table shows the three largest sub-industries (worker based) for each industry group in Appendix Table D2 at baseline. Based on Business Census Data.

D.2 Type of Maternal Labor Supply

Figure D6: Validation of Part-and Full-time Classification based on Post-Birth Earnings Share



Notes: Density of the Post-Birth Earnings Share by whether a mother reports working part-time or full-time hours in the Census. The Post-Birth Earnings Share are a mother's post-birth earnings divided by pre-birth earnings. Based on Social Security and Census data.

Table D4: Reform Impact by Type of Maternal Labor Supply

	Part-time (Earnings based)						Full-time (Earnings based)				
	Main	Controls	$< 100\%$	$< 110\%$	No trim pre-birth	No trim share	Main	Controls	$\geq 100\%$	$\geq 110\%$	No trim pre-birth
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)
Free * Treat	–0.050**	–0.050**	–0.055**	–0.063**	–0.037**	–0.052**	0.005	0.006	0.011	0.019	0.003
	(0.023)	(0.024)	(0.026)	(0.031)	(0.018)	(0.024)	(0.006)	(0.006)	(0.008)	(0.012)	(0.005)
Transition *	–0.020*	–0.021*	–0.020*	–0.025*	–0.020**	–0.021*	0.007*	0.007*	0.007	0.011	0.007*
Treat	(0.011)	(0.011)	(0.012)	(0.014)	(0.010)	(0.011)	(0.004)	(0.004)	(0.005)	(0.007)	(0.004)
Adj. R^2	0.024	0.025	0.028	0.032	0.027	0.022	0.082	0.091	0.076	0.070	0.091
Obs	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672	3,834,672
Dep mean	0.301	0.301	0.328	0.353	0.471	0.326	0.087	0.087	0.061	0.035	0.115

Notes: Estimates of Equation 2. The sample consists of wage employed mothers with a first birth after 1982 in post-birth years (see Section 5.3). Outcome is an indicator equal to one if a mother is classified as part-time (columns 1–6) or full-time (columns 7–11) based on the post-birth earnings share, i.e. the share of pre-birth earnings recovered upon returning to employment after childbirth. Column 1: Main specification, with part-time defined as recovering $< 90\%$ of pre-birth earnings. Column 2: Additional individual level controls for additional births and birth years. Column 3: Part-time defined as recovering $< 100\%$ of pre-birth earnings. Column 4: Part-time defined as recovering $< 110\%$ of pre-birth earnings. Column 5: Based on using untrimmed pre-birth earnings to construct post-birth earnings shares. Column 6: Based on untrimmed post-birth earnings share. Columns 7–11 equivalent to Columns 1–5, but for full-time indicator.

Table D5: Workforce Participation by Labor Market Attachment at Baseline

	Not in Workforce	In Workforce	Not in Wage Employment	In Wage Employment
	(1)	(2)	(3)	(4)
Free * Treat	–0.041***	–0.026***	–0.045***	–0.025***
	(0.013)	(0.008)	(0.013)	(0.008)
Transition * Treat	–0.009	–0.011*	–0.010	–0.010*
	(0.007)	(0.006)	(0.006)	(0.006)
Adj. R^2	0.145	0.049	0.121	0.049
Obs	3,191,344	4,826,848	3,585,280	4,432,912
Dep mean	0.210	0.830	0.280	0.829
P-value		.031		.004

Notes: Estimates of Equation 2 for mothers in the main sample. Outcome is an indicator for being in the workforce. Column 1: Mothers who are not in the workforce at baseline. Column 2: Mothers who are in the workforce at baseline. Column 3: Mothers who are not in wage employment at baseline. Column 4: Mothers who are in wage employment at baseline. P-value for test of equality of coefficients between columns 1 & 2, and 3 & 4 respectively. Dep mean is the pre-reform period mean of the dependent variable. All regressions include municipality and year fixed effects and individual controls measured at baseline. Standard errors clustered at the CZ level. Based on social security data. * p<0.1, ** p<0.05, *** p<0.01

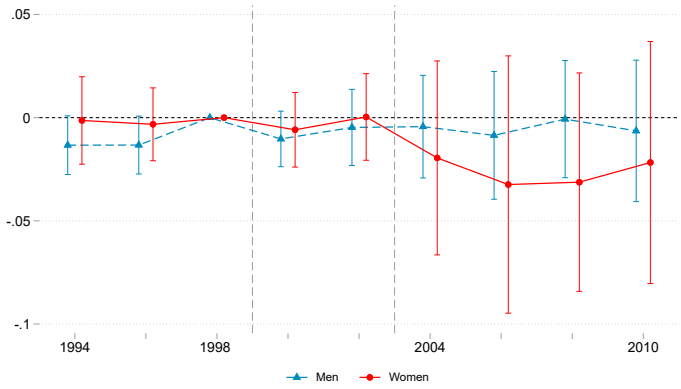
D.3 Alternative Mechanisms

Table D6: Wage Gaps between CBW and Swiss Workers

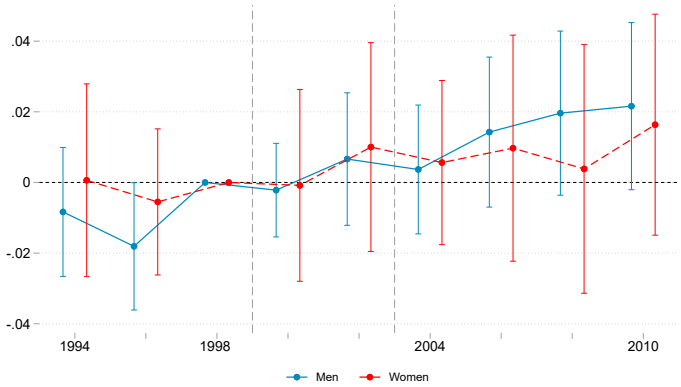
	All Natives				Female Natives				
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Cross Border	–0.056	–0.048**	–0.027***	–0.014***	0.045	0.031	0.041***	0.038***	–0.005
Worker	(0.039)	(0.020)	(0.006)	(0.005)	(0.050)	(0.030)	(0.009)	(0.004)	(0.004)
Individ. Controls		X	X	X		X	X	X	X
Female		X	X	X					X
Occ + Ind FE			X	X			X	X	X
Establishment FE				X				X	X
Adj. R^2	0.095	0.487	0.596	0.758	0.118	0.433	0.565	0.748	0.753
Obs	1,117,965	1,091,572	1,091,572	1,084,748	698,861	682,434	682,434	673,747	673,747
Dep mean	3.553	3.553	3.553	3.553	3.489	3.489	3.489	3.489	3.489

Notes: Wage gap estimates for CBW relative to Swiss workers in the post-period (2004-2010) in the border region. Outcome is the log of real hourly wages trimmed at the 1st and 99th percentile. Columns 1–4: CBW compared to all Swiss natives. Columns 5 –9: CBW compared to Swiss female workers only. Columns 1 & 5: Raw wage gap. Columns 2 & 6: Individual level controls: age FE, married, education (low, mid, high), tenure FE, management position (low, mid, high) + gender in Column 2. Columns 3 & 7: Individual controls + industry and occupation fixed effects + gender in Column 3. Columns 4 & 8: Individual controls + industry and occupation + establishment fixed effects + gender in Column 4. Column 9: Same as column 8 + gender. All specifications include year and municipality fixed effects. Robust standard errors. Data from SESS.

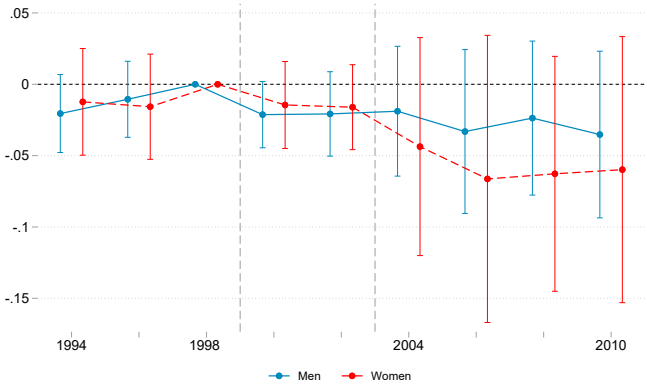
Figure D7: Reform Impact on Hourly Wages of Employed Women and Men



(a) Log Hourly Wage



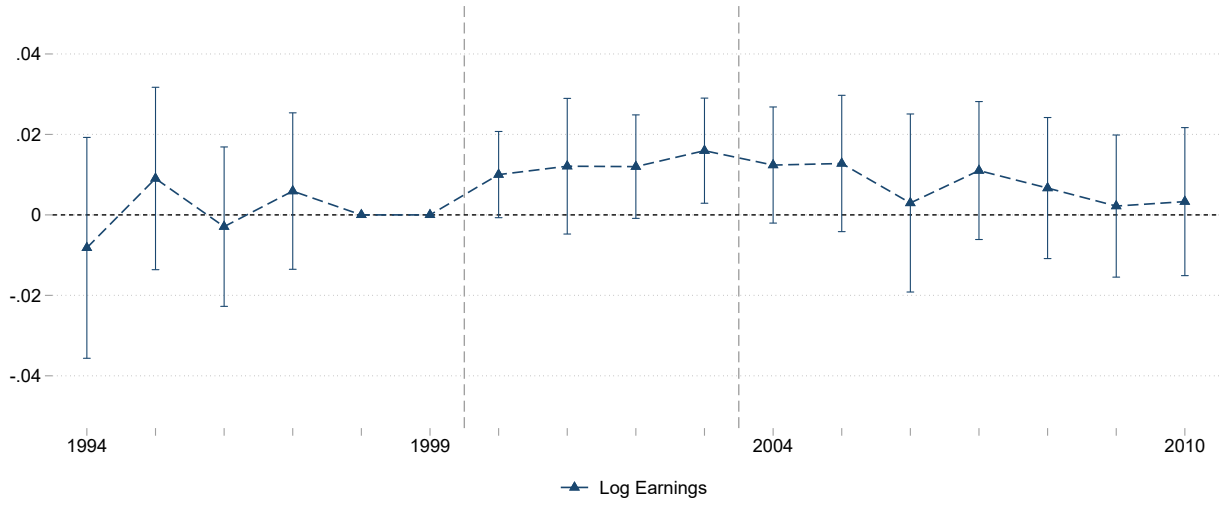
(b) Log Hourly Wage | Individual Contract



(c) Log Hourly Wage | Collective Contract

Notes: Estimates of Equation 1. The sample includes Swiss workers who are between 25 and 54 years old in 2000 on regular contracts (i.e., contracts not on an hourly basis). The outcome is the log of hourly wages, trimmed at the 1st and 99th percentile by year. Panel (a): Log hourly wage for full sample. Panel (b): Log hourly wage for workers with a contract that was negotiated on an individual basis. Panel (c): Log hourly wage for workers with a contract that was negotiated in a collective agreement. All regressions include municipality and year fixed effects and control for age, age squared and level of education (low, mid, high). Standard errors are clustered at the CZ level. Dashed vertical lines indicate the Transition (2000-2003) and Free (2004-) reform period. Based on SESS data.

Figure D8: Reform Impact on Earnings of Male Spouse



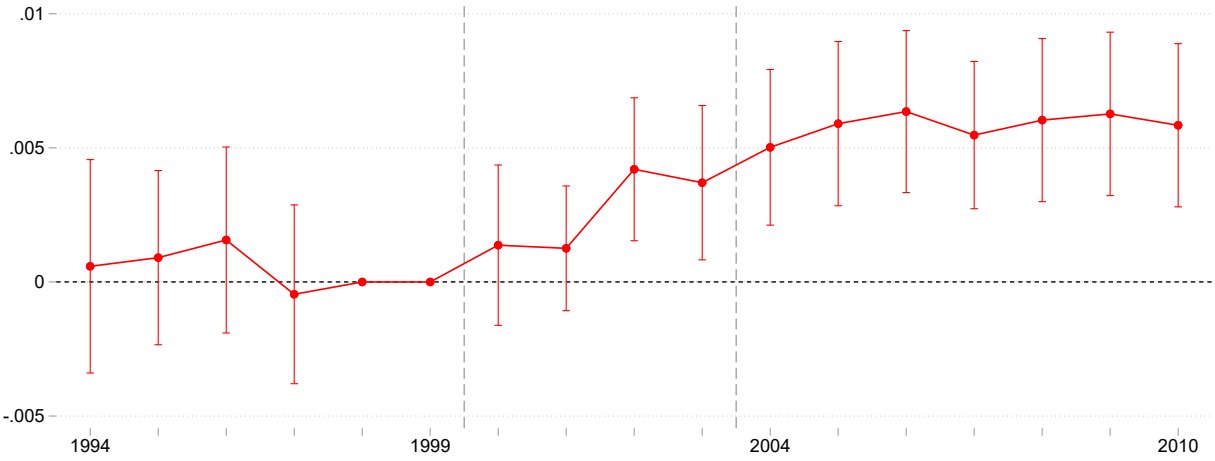
Notes: Estimates of Equation 1. Outcome is the log of earnings of the male partner for our main sample of mothers. The specification includes municipality and year fixed effects, and individual level controls measured at baseline. Standard errors are clustered at the CZ level. Based on Social Security data.

Table D7: Semi-Elasticity: Mother's Labor Supply and Partner's Earnings

	(1)	(2)	(3)	(4)
Log Partner's	–0.020***	–0.016***	–0.014***	–0.027***
Earnings	(0.003)	(0.002)	(0.003)	(0.002)
Individual FE		X	X	X
Adj. R^2	0.132	0.626	0.622	0.496
Obs	1,664,573	1,656,152	863,711	2,746,367
Dep mean	0.557	0.557	0.550	0.640

Notes: This table shows the semi-elasticity of mother's labor supply with respect to partner's earnings. It regresses an indicator for mother's workforce participation on the log of partner's earnings for different samples. Column 1: Full sample, pre-reform period only. Column 2: Full sample, pre-reform period only with individual fixed effects. Column 3: Control region only, pre-treatment period with individual fixed effects. Column 4: Control region only, all time periods (1994-2010) with individual fixed effects. Dep mean is the pre-reform period mean of the dependent variable. All regressions include municipality and year fixed effects. Standard errors clustered at the CZ level. Based on Social Security data. * p<0.1, ** p<0.05, *** p<0.01.

Figure D9: Reform Impact on Likelihood of Giving Birth



Notes: Estimates of Equation 1. Outcome is an indicator for giving birth in a given year for mothers in our main sample. The specification includes municipality and year fixed effects, and individual level controls measured at baseline. Standard errors are clustered on the CZ level. Based on Social Security data.

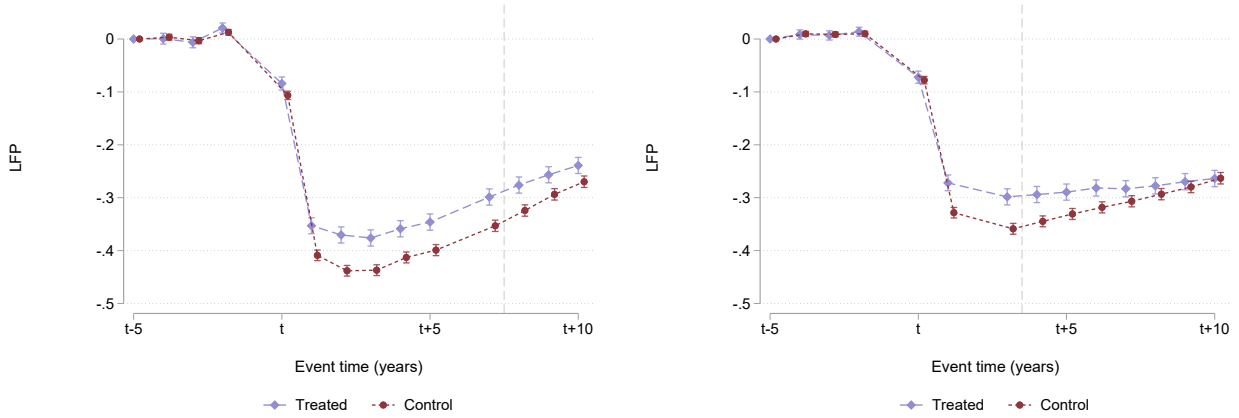
Table D8: Birth

	P(Birth)			In Workforce	
Age:	All	< Median	≥ Median	< Median	≥ Median
	(1)	(2)	(3)	(4)	(5)
Free * Treat	0.005***	0.010***	–0.002	–0.023***	–0.056***
	(0.002)	(0.003)	(0.002)	(0.007)	(0.021)
Transition *	0.002	0.005	–0.001	0.007	–0.031**
Treat	(0.001)	(0.003)	(0.001)	(0.009)	(0.013)
Adj. R^2	0.092	0.086	0.085	0.067	0.094
Obs	7,933,936	3,757,808	4,176,128	3,790,176	4,228,016
Dep mean	0.104	0.167	0.047	0.634	0.538

Notes: Estimates of Equation 2 for mothers in our main sample. Outcome is an indicator for giving birth (columns 1–3) and an indicator for being in the workforce (columns 4–5). Column 1: Full sample. Columns 2 and 4: Mothers below median age at baseline. Columns 3 and 5: Mothers at or above median age at baseline. Dep mean is the pre-reform period mean of the dependent variable. All regressions include municipality and year fixed effects, and individual level controls measured at baseline. Standard errors clustered at the CZ level. Based on Social Security data. * p<0.1, ** p<0.05, *** p<0.01.

E Child Penalty Estimates

Figure E1: Child Penalty in Participation



(a) Birth Cohort 1992

(b) Birth Cohort 1996

Notes: Child penalty in participation for mothers with a first birth in 1992 and 1996, living in treated municipalities (Treated) or control municipalities (Control) at baseline. Vertical gray lines indicate start of the post-reform period. Observations in the spillover region are excluded. Based on Social Security data.

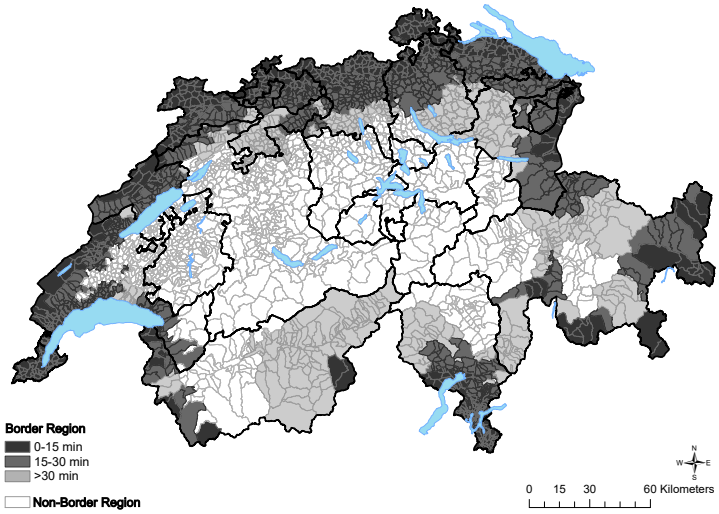
Table E1: Reform Impact on the Child Penalty in Participation

	Full Sample		Omit 1998	
	(1)	(2)	(3)	(4)
Post * Border * T = 0	–0.012	–0.012	–0.011	–0.010
	(0.008)	(0.008)	(0.008)	(0.008)
Post * Border * T = 1	–0.023**	–0.022**	–0.017*	–0.017
	(0.010)	(0.010)	(0.010)	(0.010)
Post * Border * T = 2	–0.019**	–0.019**	–0.018**	–0.018**
	(0.008)	(0.008)	(0.008)	(0.008)
Post * Border * T = 3	–0.022***	–0.022***	–0.019**	–0.019**
	(0.008)	(0.008)	(0.008)	(0.008)
Post * Border * T = 4	–0.018***	–0.018***	–0.014**	–0.014**
	(0.007)	(0.007)	(0.006)	(0.006)
Post * Border * T = 5	–0.019***	–0.019***	–0.015**	–0.015**
	(0.007)	(0.007)	(0.006)	(0.006)
Post * Border * T = 6	–0.016***	–0.016**	–0.011*	–0.011*
	(0.006)	(0.006)	(0.006)	(0.006)
Post * Border * T = 7	–0.016**	–0.016**	–0.013*	–0.013*
	(0.008)	(0.008)	(0.007)	(0.007)
Post * Border * T = 8	–0.007	–0.007	0.004	0.004
	(0.007)	(0.007)	(0.007)	(0.007)
Post * Border * T = 9	–0.018**	–0.018**	–0.002	–0.002
	(0.008)	(0.008)	(0.007)	(0.007)
Post * Border * T = 10	–0.014	–0.014	–0.012	–0.012
	(0.010)	(0.010)	(0.010)	(0.010)
Mother's Age FE	X		X	
Adj. R^2	0.130	0.129	0.133	0.133
Obs	2,803,472	2,803,472	2,433,540	2,433,540
Dep mean	–0.290	–0.290	–0.290	–0.290

Notes: This table shows the post-reform change in the child penalty in labor force participation for women with a first birth between 1989 and 2000. Outcome is an indicator for labor force participation. Columns 1 and 3 report estimates for equation 3 that includes age fixed effects and assumes no anticipation in $t = -5$, following Thakral and Tô (2026). Columns 2 and 4 assume age effects to be zero. Column 1 & 2: Full sample. Column 3 & 4: Excludes social security data from the year 1998 and the 1999 birth cohort, as 1998 constitutes the baseline year for this cohort (see Section 3). The dependent mean reports the average child penalty in participation for the control group ($t = 0$ – $t = 10$) Observations in the spillover region excluded from the sample. Based on Social Security data. * p<0.1, ** p<0.05, *** p<0.01.

F Further Reform Context

Figure F1: The border and non-border region and travel distance to the border



Notes: This figure depicts municipalities in the border region in three different shades of grey and those in the non-border region in white. Within the border region, municipalities are differentiated according to their travel time by car to the nearest border crossing. The black lines denote cantonal borders. Reproduced from Beerli et al. (2021).